A fully Lean-verified quadratic lower bound on the border determinantal complexity of $x_1^n + \cdots + x_n^n$ over $\mathbb{C}$, including the symmetric and exact models

Proof by conormal specialization

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We present the first end-to-end Lean-verified superlinear lower bound on border determinantal complexity. Let $F_n = x_1^n + \cdots + x_n^n$. We prove, over $\mathbb{C}$, that

$$\overline{\mathrm{dc}}(F_n) \geq \frac{(n-1)^2}{4e}, \qquad \overline{\mathrm{sdc}}(F_n) \geq \frac{(n-1)^2}{2e},$$

for the ordinary and symmetric border determinantal models, and therefore the same bounds for exact determinantal complexity. Together with known $O(n^2)$ constructions, both border quantities are $\Theta(n^2)$.

The proof compares one enumerative invariant on two sides of a degeneration. On the determinant side, the $(n-2)$nd multidegree of the multiplicity-one Gauss graph of any size-m affine-linear determinant is bounded by an explicit multihomogeneous Bézout number; no smoothness, irreducibility, or squarefreeness is assumed. On the target side, a flat limit of these effective Gauss-graph cycles contains the conormal variety of the projective Fermat cone. Its relevant multidegree is the classical dual degree $n(n-1)^{n-2}$. Effectivity gives the inequality in the correct direction, and an $(n-1)$st root yields the quadratic bound.

All four bounds are fully verified in Lean 4 with Mathlib. The Lean theorems assume only $n \geq 3$ and a literal determinantal representation or literal border membership, and they depend on no axioms beyond Lean's foundational ones. The formal proof follows a localized version of the argument presented here: where the text compares global cycles, it compares local intersection lengths of finite slices of those cycles.

Lean sources: https://github.com/karthik202321/power-sum-dc-lean.
Archived release: https://doi.org/10.5281/zenodo.23274251.

The exposition begins with a geometric dictionary, a one-line proof chain, and a dependency diagram; the longer incidence and flatness arguments then verify those high-level steps.

Additional Key Words and Phrases: border determinantal complexity, algebraic complexity, conormal variety, polar degree, Gauss map, geometric complexity theory, Lean, formal verification

How the paper is organized

The paper is written in three layers. The first layer — the Introduction and Section 2 — explains the whole argument without requiring scheme theory or intersection theory. The second layer consists of the blue, green, and gold boxes placed immediately before the technical proofs; each box states the goal of the next calculation, the input it uses, and the output that will be needed later. The third layer is the complete written proof. The boxes are explanatory only: every mathematical implication is also proved in the ordinary theorem–lemma text. A fourth layer, Appendix D, reports the complete Lean verification of the exact and border numerical endpoints and explains how to reproduce it independently.

1 Introduction

1.1 The question and the answer

For a polynomial $f(x_1, \ldots, x_n)$, its determinantal complexity is the smallest m for which

$$f(x) = \det(A_0 + x_1 A_1 + \cdots + x_n A_n)$$

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with constant $m \times m$ matrices A_i. Let

$$D_m = \left\{ \det\left(A_0 + \sum_{i=1}^{n} x_i A_i \right) : A_i \in \mathbb{C}^{m \times m} \right\}.$$

The *border determinantal complexity* is

$$\overline{\mathrm{dc}}(f) = \min\{m : f \in \overline{D_m}\},$$

where the closure is Zariski closure (equivalently, over $\mathbb{C}$, Euclidean closure for this constructible set). Thus f need only be a limit of size-m determinants. Restricting the matrices to be symmetric defines $\overline{\mathrm{sdc}}(f)$.

We study the diagonal power sum

$$F_n = x_1^n + \cdots + x_n^n.$$

Exact determinantal lower bounds for explicit polynomials have been stubbornly close to linear in the number of variables, while the standard construction for F_n has quadratic size. The following result closes that gap up to a constant even in the border model.

Theorem 1.1 (Main theorem: ordinary model). *For every $n \geq 3$, over $\mathbb{C}$,*

$$\overline{\mathrm{dc}}(F_n) \;\geq\; \frac{(n-1)^2}{4e}.$$

Theorem 1.2 (Main theorem: symmetric model). *For every $n \geq 3$, over $\mathbb{C}$,*

$$\overline{\mathrm{sdc}}(F_n) \;\geq\; \frac{(n-1)^2}{2e}.$$

Known constructions give $\mathrm{dc}(F_n) = O(n^2)$ and $\mathrm{sdc}(F_n) = O(n^2)$ [7, 14]. Since border complexity is at most exact complexity, the two theorems imply that both border quantities are $\Theta(n^2)$.

Corollary 1.3 (Exact bounds). *For every $n \geq 3$,*

$$\mathrm{dc}(F_n) \geq \frac{(n-1)^2}{4e}, \qquad \mathrm{sdc}(F_n) \geq \frac{(n-1)^2}{2e}.$$

1.2 End-to-end Lean verification

Theorems 1.1 and 1.2 and Corollary 1.3 are verified end to end in Lean 4 with Mathlib. To our knowledge, they are the first determinantal or border determinantal lower bounds verified end to end in a proof assistant. The Lean statements (Appendix D.1) assume only $n \geq 3$ and either a literal determinantal representation of F_n or literal membership of F_n in the closure of the size-m determinants. They depend on no axioms beyond Lean's three foundational ones (`propext`, `Classical.choice` and `Quot.sound`). Every intermediate input—Bézout's theorem for isolated zeros with excess, the generic flags, the degree computation and the one-parameter family—is constructed or proved inside the development, not assumed.

The Lean proof follows a localized version of the argument in this paper. The text compares global cycles: the conormal variety of the Fermat cone and the limit of the Gauss-graph cycles. The Lean proof compares finite slices of them. A *local Gauss section* of a hypersurface $f = 0$ is the finite scheme cut out by $f = 0$, the $n-2$ polar equations $\sum_i \ell_{t,i}\, \partial_i f = 0$ $(1 \leq t \leq n-2)$ and one affine chart equation $a \cdot x = 1$. It is the zero-dimensional slice of the Gauss graph that the multidegree δ_{n-2} counts (Section 2.3), and the lengths of its local rings are the multiplicities of its points. Comparing these lengths across the degeneration gives the inequalities of (1) in local form. The verification therefore covers the stated theorems and every lemma in their dependency

chain, but not every sentence about global cycles in the exposition. Appendix D gives the exact Lean statements, their correspondence with the argument, and build and audit instructions.

Formalization and archival links

Lean sources: https://github.com/karthik202321/power-sum-dc-lean.
Archived release (Lean sources, audits and this paper): https://doi.org/10.5281/zenodo.23274251.
arXiv record of this paper: https://arxiv.org/abs/2606.13628.

1.3 The proof in one displayed line

Let $X \subset \mathbb{P}^n$ be the projective cone defined by the same form F_n (now viewed as independent of x_0). If $F_n \in \overline{D_m}$, that is, if F_n is a limit of size-m determinants, we construct a nearby determinantal fibre with multiplicity-one Gauss-graph cycle Γ_c. The proof is the chain

$$\underbrace{n(n-1)^{n-2}}_{\text{Fermat dual degree}} = \delta_{n-2}(\mathrm{Con}(X)) \le \underbrace{\delta_{n-2}(\Gamma_c)}_{\text{conserved effective cycle}} \le \underbrace{B(m,n)}_{\text{kernel incidence}} \le \left(\frac{4em}{n-1}\right)^{n-1}. \quad (1)$$

The first equality is classical projective duality. The first inequality is the specialization step: the conormal variety of the Fermat cone occurs as an effective component of the limiting Gauss-graph cycle. The second inequality is a determinant-specific Bézout count. The final inequality is elementary. Since the left side is at least $(n-1)^{n-1}$, taking $(n-1)$st roots gives $m \ge (n-1)^2/(4e)$. In the symmetric model the determinant-side estimate improves by a factor of two in the base of the power.

Roadmap: the five proof obligations behind (1)

(1) **Turn border membership into a family.** From $F_n \in \overline{D_m}$, construct a curve of polynomials G_c with $G_0 = F_n$ and with every generic G_c an exact size-m determinant. This is Lemma 5.1.

(2) **Package tangent data in a flat family.** For each generic fibre, form the multiplicity-one Gauss graph Γ_c. Keep the components that move with the parameter and close them up to a proper flat family $W \to C$. This is Section 7.

(3) **Show that the target survives.** Prove that the conormal variety of the Fermat cone is an irreducible component of the effective special fibre $[W_0]$. This is Proposition 8.1.

(4) **Compute the target-side count.** Show that the relevant multidegree of the Fermat cone is $n(n-1)^{n-2}$. This is Proposition 9.1.

(5) **Bound the determinant-side count.** Lift generic Gauss points to left and right kernel lines and count the lifts by a multihomogeneous Bézout number. This is Theorem 1.4.

Once these five statements are available, the main theorem is the arithmetic in Section 10.

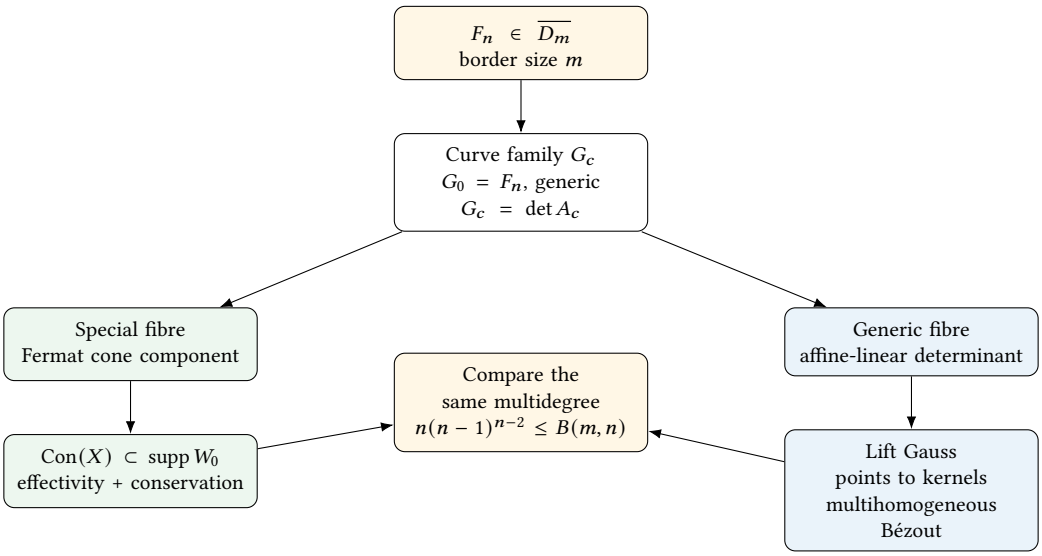


Fig. 1. The proof separates into a target-side lower count and a determinant-side upper count. Flat specialization lets the two counts be compared.

1.4 The two counts and the bridge between them

The proof compares a *lower count* attached to the target polynomial with an *upper count* attached to any size-m determinant. The quantities are deliberately chosen to be the same multidegree.

	Target side	Determinant side
Geometric object	Conormal variety of the Fermat cone	Multiplicity-one Gauss graph of $\det \widehat{A}$
Quantity counted	δ_{n-2}, equal to $n(n-1)^{n-2}$	The same δ_{n-2}, at most $B(m,n)$
Reason for the bound	Classical dual degree of a smooth Fermat hypersurface	Kernel lift plus multihomogeneous Bézout
Bridge	A flat effective limit contains the target conormal, so target count $\le$ generic count.	

The central design choice is therefore not merely to find a large invariant of F_n, but to find one that can be compared across a degeneration without changing sign or losing multiplicity.

1.5 Why the border step is not automatic

The exact-representation argument reads a polar or conormal degree from a fixed determinant. Such degrees are not closed conditions on polynomial space: under degeneration, isolated solutions can merge into singular strata or excess components. A local singularity invariant at the vertex of the Fermat cone can even jump *up* in the special fibre, so the naive semicontinuity direction is wrong.

The repair is to specialize a *global effective cycle*, not a local number. The Gauss graphs of the generic fibres have a flat limit W_0. Their multidegrees are conserved, and every irreducible component of W_0 has a nonnegative coefficient. Once $\mathrm{Con}(X)$ is shown to be one of those components, its multidegree is automatically at most the conserved total. Extra components are harmless because they add rather than subtract.

A second issue is non-reducedness. The degree-m homogenization of the special fibre is $x_0^{m-n}F_n$, and nearby determinants may also have repeated factors. We therefore use the *multiplicity-one Gauss graph* $\Gamma_1(F)$: it is the sum of the conormal varieties of those irreducible components on which

F vanishes with multiplicity exactly one. Equivalently, it is the part of the hypersurface seen by the nonzero differential dF. This is both the cycle controlled by the determinant-side incidence and the cycle that captures the Fermat component under specialization.

1.6 The determinant-side estimate

The only long algebraic calculation in the proof is isolated in the following theorem. Its role is simple: a size-m determinant cannot have an arbitrarily large $(n-2)$nd conormal multidegree.

Theorem 1.4 (Determinantal conormal bound). *Let $n \geq 2$, $m \geq 1$, and let $A(x) = A_0 + \sum_{i=1}^{n} x_i A_i$ be an $m \times m$ matrix of affine-linear forms over $\mathbb{C}$. Let $\widehat{A} = x_0 A_0 + \sum_i x_i A_i$ and suppose $F = \det \widehat{A} \not\equiv 0$. Then*

(i)

$$\delta_{n-2}(\Gamma_1(F)) \leq B(m,n),$$

where

$$B(m,n) = \left[x^n u^{m-1} v^{m-1}\right] x(x+u)^m (x+v)^{m-1} (u+v)^{n-2}$$

and equivalently

$$B(m,n) = \sum_{i=1}^{n-1} \binom{m}{i} \binom{m-1}{n-1-i} \binom{n-2}{i-1}.$$

(ii) If all A_i are symmetric, then

$$\delta_{n-2}(\Gamma_1(F)) \leq 2^{n-2} \binom{m}{n-1}.$$

No smoothness, irreducibility, squarefreeness, or genericity assumption is made on F or on the matrices A_i.

What to remember from Theorem 1.4

The exact coefficient formula is needed only at the end. Conceptually,

$$B(m,n) \leq \left(\frac{4em}{n-1}\right)^{n-1}, \qquad 2^{n-2}\binom{m}{n-1} \leq \left(\frac{2em}{n-1}\right)^{n-1}.$$

The proof lifts each Gauss point to kernel vectors and counts the lifted points by Bézout.

1.7 Context and relation to prior work

Determinantal complexity is a central model in algebraic complexity, and border complexity is the natural orbit-closure version used in geometric complexity theory. For exact determinantal complexity of F_n, the prior published lower bound was $1.5n - 3$ [7], following earlier singular-locus methods [1]. Mignon and Ressayre's quadratic lower bound for the permanent [10] extends to the border through a dual-*dimension* obstruction [9]; measured in the permanent's number of variables, that bound is linear. To our knowledge, Theorems 1.1 and 1.2 give the first border determinantal lower bounds for an explicit n-variate family that are superlinear in n.

The paper supersedes the exact lower bounds in the companion manuscripts [13, 14].[1] The determinant-side incidence originates there, but Theorem 1.4 removes their smoothness requirement. The new ingredient is the effective conormal-cycle specialization that transfers the lower bound to border complexity.

[1] A shorter proof of a quadratic bound in the exact ordinary model was later given in [8]; by its authors' account, its outline came out of parsing parts of the proof in an earlier version of this paper. It is cited here as a reading aid for that case only. No argument in this paper uses it, and it does not treat the border or symmetric models.

1.8 How to read the paper

Section 2 gives a geometric dictionary and a concrete low-dimensional picture. Section 3 records the formal definitions. Section 4 proves the determinant-side black box; it may be skipped on a first reading. Sections 5–8 construct and analyze the degeneration, Section 9 computes the target-side multidegree, and Section 10 combines the inequalities. Readers who want only what is machine-checked can read Section 1.2 and Appendix D.1.

2 A geometric guide for TCS readers

This section translates the intersection-theoretic language into the finite counts used by the proof. Nothing in this section replaces a later proof; its purpose is to make the objects and the direction of every inequality familiar before the technical construction begins.

2.1 Why projective space and homogenization appear

An affine size-m determinant has degree at most m:

$$G(x) = \det\left(A_0 + \sum_{i=1}^{n} x_i A_i\right).$$

Introduce a homogenizing variable x_0 and set

$$\widehat{A}(x_0, x) = x_0 A_0 + \sum_{i=1}^{n} x_i A_i.$$

Then $\det \widehat{A}$ is a homogeneous degree-m form, and on the affine chart $x_0 = 1$ it equals G. Passing to projective space has three concrete advantages.

(1) Points that move to infinity remain inside the compact projective ambient space; properness later guarantees that limiting Gauss points do not simply disappear.
(2) Tangent hyperplanes themselves form a projective space $(\mathbb{P}^n)^\vee$, so a point and its tangent hyperplane naturally live in the product $\mathbb{P}^n \times (\mathbb{P}^n)^\vee$.
(3) Intersection numbers with generic linear spaces are finite and stable; these are the multidegrees used in the proof.

The target F_n has degree n, whereas the approximating determinant has formal homogeneous degree m. Homogenizing at degree m therefore gives

$$\widehat{G}_0 = x_0^{m-n} \widetilde{F}_n.$$

The factor x_0^{m-n} is the padding needed to compare degree-m forms. It may make the special divisor non-reduced at infinity, but it does not increase the multiplicity of the Fermat factor itself. This distinction is used throughout the specialization argument.

2.2 Gauss graphs and conormal varieties

Let $S = V(f) \subset \mathbb{P}^n$ be a smooth hypersurface. At a point $x \in S$, the gradient $df(x)$ defines the tangent hyperplane. The map

$$x \longmapsto (x, [df(x)])$$

places each point together with its tangent hyperplane in $\mathbb{P}^n \times (\mathbb{P}^n)^\vee$. The closure of this graph is the conormal variety $\mathrm{Con}(S)$.

If f is reducible or non-reduced, the differential vanishes identically on any component that occurs with multiplicity at least two. The cycle $\Gamma_1(f)$ keeps precisely the conormal varieties of

the multiplicity-one components. This is not an arbitrary convention: it is exactly the part of the divisor visible in the open set where $df \neq 0$.

There are two useful elementary examples.

Example 2.1 (One simple component and one repeated component)**.** Let $F = gh^2$, with g and h distinct reduced irreducible forms. At a generic point of $V(g)$ away from $V(h)$,

$$dF = h^2\, dg,$$

which is nonzero, so the conormal of $V(g)$ is present in $\Gamma_1(F)$. At a generic point of $V(h)$, every term in dF contains a factor h, so $dF = 0$; the repeated component $V(h)$ is absent. Thus $\Gamma_1(F) = \mathrm{Con}(V(g))$.

Example 2.2 (Intersections do not remove multiplicity-one components)**.** Let $F = gh$ with g, h reduced and distinct. At a generic point of $V(g)$ away from $V(h)$, $dF = h\, dg \neq 0$, and similarly on $V(h)$. At the intersection $V(g) \cap V(h)$ the differential may vanish, but that lower- dimensional set does not matter: taking the closure of the open Gauss graph recovers both conormal varieties. Hence $\Gamma_1(F) = \mathrm{Con}(V(g)) + \mathrm{Con}(V(h))$ as a cycle.

The word *cycle* is important. It means a formal sum of irreducible subvarieties with integer coefficients. Here every visible conormal component is assigned coefficient one. Later, a special fibre can contain the same component with a larger positive coefficient; effectivity ensures that such multiplicities help rather than hurt the desired inequality.

2.3 What a multidegree counts

An $(n-1)$-dimensional cycle $C \subset \mathbb{P}^n \times (\mathbb{P}^n)^\vee$ has numbers

$$\delta_k(C) = \deg\left(C \cdot h_1^{n-1-k} h_2^k\right), \qquad 0 \leq k \leq n-1.$$

Enumeratively, $\delta_k(C)$ is the number of points of C satisfying $n-1-k$ generic linear conditions on the primal point and k generic linear conditions on the dual hyperplane. The dimension check is exact: C has dimension $n-1$, and the total number of imposed conditions is $(n-1-k)+k = n-1$. For generic choices the intersection is therefore zero-dimensional, transverse, and supported away from any prescribed lower- dimensional bad locus.

We use only $k = n-2$. Thus a counted point must satisfy

(1) one generic linear condition on the point $x \in \mathbb{P}^n$; and
(2) $n-2$ generic linear conditions on its tangent hyperplane $[df(x)] \in (\mathbb{P}^n)^\vee$.

For $n = 3$, this is especially concrete: one plane condition on the point and one hyperplane condition on the tangent plane.

The adjective *generic* does not mean that the determinant or the family is generic. The input matrix may be completely special. Only the auxiliary linear spaces used to read the intersection number are chosen after the input, outside a proper closed set of bad choices.

Why multidegrees are suitable for specialization

They measure the whole fibre cycle. In the proper flat family inside our fixed biprojective ambient space, every multidegree is conserved. If the special fibre splits into several effective components, each component contributes a nonnegative amount to every multidegree.

2.4 Effective cycles and flat families in plain language

An effective cycle has the form

$$a_1 V_1 + \cdots + a_r V_r, \qquad a_i \in \mathbb{Z}_{\geq 0}.$$

Every multidegree is additive: $\delta_k(\sum_i a_i V_i) = \sum_i a_i \delta_k(V_i)$, and each summand is nonnegative. Consequently, if a special fibre contains a desired component V_1 with coefficient at least one, then

$$\delta_k(V_1) \le \delta_k\left(\sum_i a_i V_i\right).$$

This elementary inequality is the sign mechanism of the border proof.

Flatness is the geometric condition that makes the *total* fibre cycle specialize without losing intersection number. A useful one-variable analogy is a polynomial whose roots move with a parameter. Two simple roots may coalesce into a double root, so the number of distinct roots drops, but the number counted with multiplicity is conserved. In higher-dimensional families, the extra multiplicity may be carried by an entire component over a singular stratum; the cycle formalism records it automatically.

2.5 A model degeneration: where the missing count goes

For a smooth plane cubic, the dual curve has degree six. When the cubic degenerates to a nodal cubic, the reduced conormal contributes only four generic tangencies; the remaining multiplicity two is supported over the node. The total specialized cycle still has degree six:

$$6 = 4 + 2.$$

This is the picture to keep in mind. The visible conormal of the special hypersurface can have smaller degree than a generic fibre, but the deficit reappears as effective excess. Our proof only needs the inequality "one special component $\le$ conserved total."

2.6 A fully worked low-dimensional picture: $n = 3$

Take $F_3 = x^3 + y^3 + z^3$. Its projective zero set $Z \subset \mathbb{P}^2$ is a smooth plane cubic, whose dual curve has degree $3(3-1) = 6$. The projective cone $X \subset \mathbb{P}^3$ over Z has ruling lines joining the vertex to the points of Z. The relevant invariant is $\delta_1(\mathrm{Con}(X))$:

- one generic condition is imposed on the point of the cone;
- one generic condition is imposed on the tangent plane;
- the tangent-plane condition selects six tangencies on the base cubic;
- the point condition then cuts the corresponding six ruling lines once.

Hence $\delta_1(\mathrm{Con}(X)) = 6$.

On the determinant side the general coefficient simplifies to

$$B(m,3) = 3\binom{m}{2}.$$

Thus the exact comparison in this small case reads $6 \le 3\binom{m}{2}$, which forces $m \ge 3$. This is not the asymptotic content of the theorem, but it displays every geometric ingredient without high-dimensional notation. Under a nodal degeneration, the same total six may split as four ordinary tangencies plus multiplicity two over the node: $6 = 4 + 2$.

2.7 Why the Fermat cone has a large count

Let

$$Z = V(x_1^n + \cdots + x_n^n) \subset \mathbb{P}^{n-1}$$

be the smooth Fermat hypersurface, and let $X \subset \mathbb{P}^n$ be its cone with vertex $q_0 = [1 : 0 : \cdots : 0]$. Every tangent hyperplane to X contains q_0. Therefore the top dual multidegree $\delta_{n-1}(\mathrm{Con}(X))$

vanishes. In the next slot, a generic primal hyperplane meets each ruling line of the cone once, leaving the dual-degree count of the base:

$$\delta_{n-2}(\mathrm{Con}(X)) = \deg Z^{\vee} = n(n-1)^{n-2}.$$

The last equality is the classical dual-degree formula for a smooth degree-n hypersurface in $\mathbb{P}^{n-1}$.

2.8 Why determinants admit an upper count

At a smooth point of the determinant hypersurface where $\widehat{A}(x)$ has corank one, choose left and right kernel vectors u and v. The adjugate has rank one,

$$\operatorname{adj} \widehat{A}(x) = \alpha\, vu^{\mathsf{T}},$$

and Jacobi's formula gives

$$\partial_i \det \widehat{A}(x) = \alpha\, u^{\mathsf{T}} \widehat{A}_i v.$$

Thus the tangent hyperplane is encoded by the bilinear kernel pairings $u^{\mathsf{T}}\widehat{A}_i v$. A Gauss-graph point can therefore be lifted to $(x, [u], [v])$. After adding the generic flag conditions defining δ_{n-2}, these lifts satisfy a square multihomogeneous system. Theorem 1.4 proves that the desired lifts are isolated and then applies Bézout.

The determinant-side proof can be read as the following six-step algorithm.

(1) Use a generic flag to turn the desired multidegree into finitely many smooth Gauss points $(x, [dF(x)])$.
(2) At each point, use $dF(x) \neq 0$ to deduce $\operatorname{rank} \widehat{A}(x) = m - 1$ and therefore unique projective left and right kernel lines $[u]$ and $[v]$.
(3) Express the tangent covector through the kernel pairings $u^{\mathsf{T}}\widehat{A}_i v$.
(4) Write all kernel and flag conditions as a square system on $\mathbb{P}^n \times \mathbb{P}^{m-1} \times \mathbb{P}^{m-1}$.
(5) Prove locally, by a Schur-complement calculation, that every desired lift is an isolated reduced solution of that system.
(6) Apply multihomogeneous Bézout. Other solutions or excess components may exist, but they only consume nonnegative intersection multiplicity.

2.9 Dictionary of the proof

Object	Plain-language meaning	Role in the proof
$\overline{\mathrm{dc}}(f)$	Smallest determinant size that can approximate f	Starting hypothesis
$\Gamma_1(F)$	Tangent-hyperplane graph of multiplicity-one components	Survives repeated factors
δ_{n-2}	One condition on the point, $n-2$ on its tangent hyperplane	Common quantity compared on both sides
W_0	Effective flat limit of nearby Gauss graphs	Conserves the total count
$\mathrm{Con}(X)$	Tangent data of the Fermat cone	Supplies the large lower count
Kernel incidence	Equations for x, left kernel u, right kernel v	Supplies the determinant-side upper count
Dominates C	A component maps densely onto the parameter curve	Keeps only geometry that occurs for moving fibres
Vertical component	A component supported over finitely many parameter values	Removed before taking the moving closure
Flat over C	Fibre cycles specialize with multiplicity bookkeeping	Makes multidegrees constant
Reduced isolated point	Local solution ring has length one	Ensures one counted Gauss point contributes exactly one

3 Formal setup and reusable facts

How to use this section

This section fixes the cycle-theoretic language used in the proof. On a first reading, the essential points are: (i) $\Gamma_1(F)$ keeps exactly the components on which the differential is generically nonzero; (ii) δ_k is an enumerative count; and (iii) multidegrees of effective cycles are nonnegative.

Notation at a glance

F_n	$x_1^n + \cdots + x_n^n$, the affine target polynomial.
$\widetilde{F}_n$	The same degree-n form in $x_0, x_1, \ldots, x_n$, independent of x_0.
X	The projective cone $V(\widetilde{F}_n) \subset \mathbb{P}^n$.
$\mathrm{Con}(S)$	Closure of pairs consisting of a smooth point of S and a hyperplane tangent there.
$\Gamma_1(F)$	Sum of conormals of the multiplicity-one components of $V(F)$.
δ_k	Multidegree imposing $n-1-k$ primal and k dual linear conditions.
$G_c, \widehat{G}_c$	The affine family and its degree-m homogenization.
Γ_c	$\Gamma_1(\widehat{G}_c)$ on a generic fibre.
W	The closed flat family formed from the moving components of the open Gauss graph.

We work over $\mathbb{C}$ throughout; $n \geq 2$ is the number of affine variables, and $F_n = x_1^n + \cdots + x_n^n$. We write $\widetilde{F}_n$ for the same polynomial regarded as a degree-n form in $(x_0, x_1, \ldots, x_n)$ (it does not involve x_0), and

$$X \;:=\; V(\widetilde{F}_n) \;\subset\; \mathbb{P}^n,$$

the projective cone over the smooth Fermat hypersurface $Z = V(F_n) \subset \mathbb{P}^{n-1} = \{x_0 = 0\}$ with vertex $q_0 = [1 : 0 : \cdots : 0]$.

3.1 Conormal cycles and multidegrees

Let h_1, h_2 denote the pullbacks to $\mathbb{P}^n \times (\mathbb{P}^n)^\vee$ of the hyperplane classes of the two factors. For an $(n-1)$-cycle C on $\mathbb{P}^n \times (\mathbb{P}^n)^\vee$ and $0 \le k \le n-1$, the k-th *multidegree* is

$$\delta_k(C) \;:=\; \deg\left(C \cdot h_1^{n-1-k} h_2^{k}\right) \in \mathbb{Z}.$$

Multidegrees are additive in C and, by Lemma 3.5 below, nonnegative on effective cycles: δ_k of an irreducible $(n-1)$-fold is computed by a generic transverse flag intersection and is the (nonnegative) number of intersection points.

For an irreducible subvariety $S \subset \mathbb{P}^n$, $\mathrm{Con}(S) \subset \mathbb{P}^n \times (\mathbb{P}^n)^\vee$ is the conormal variety: the closure of $\{(x, [\xi]) : x \in S_{\mathrm{sm}},\ \xi|_{T_x S} = 0\}$. For a hypersurface S it has dimension $n-1$, and over a smooth point of $S = V(g)$ (g reduced) the fibre is the single covector $[dg(x)]$.

Definition 3.1 (Multiplicity-one Gauss-graph cycle). Let F be a nonzero form on $\mathbb{P}^n$ of degree ≥ 1, with factorization $F = \prod_j g_j^{k_j}$ into pairwise distinct irreducibles, $S_j = V(g_j)$. Set

$$\Gamma_1(F) \;:=\; \overline{\left\{(x, [dF(x)]) \;:\; F(x) = 0,\ dF(x) \neq 0\right\}} \quad \text{(closure, with the reduced structure)},$$

regarded as the $(n-1)$-cycle given by its irreducible components with coefficient one.

Lemma 3.2 (Structure of Γ_1). *$\Gamma_1(F) = \sum_{j:k_j=1} \mathrm{Con}(S_j)$ as cycles. In particular $\Gamma_1(F)$ is effective of pure dimension $n-1$ (or the zero cycle, when no component has multiplicity one), and if F is squarefree then $\Gamma_1(F)$ is the full reduced conormal cycle of $V(F)$.*

Why the factorization immediately identifies $\Gamma_1(F)$

At a generic point of one component S_j, all other irreducible factors are units. If g_j occurs once, dF is a nonzero unit times dg_j; if it occurs at least twice, every derivative still contains a factor g_j and vanishes on S_j. The proof below is exactly this calculation.

Proof. On a component of multiplicity $k_j \ge 2$ one has $dF = g_j^{k_j-1}\left(k_j(\prod_{l\neq j} g_l^{k_l})\, dg_j + g_j(\cdots)\right)$, which vanishes identically on S_j; so the open graph in Definition 3.1 contains no point lying *only* on multiplicity-≥ 2 components. Conversely, at a point x lying on S_j with $k_j = 1$ and on no other component, $dF(x) = \left(\prod_{l\neq j} g_l^{k_l}(x)\right) dg_j(x)$, a nonzero multiple of $dg_j(x)$ wherever $dg_j(x) \neq 0$; such x are dense in S_j (g_j irreducible, hence reduced), and there $[dF(x)] = [dg_j(x)]$ is the conormal covector of S_j. Hence the open graph is a disjoint union of dense opens of the $\mathrm{Con}(S_j)$ with $k_j = 1$, and its closure is their union. Each $\mathrm{Con}(S_j)$ is irreducible of dimension $n-1$. □

Example 3.1 (A factorization sanity check)**.** For $F = g^2 h^3 k$ with distinct reduced irreducible factors, the only multiplicity-one component is $V(k)$. Lemma 3.2 gives $\Gamma_1(F) = \mathrm{Con}(V(k))$, regardless of the singularities or intersections of the three components. This example is useful later when the homogenized special fibre contains a high power of x_0 together with one Fermat factor.

Example 3.2 (Tangency: multiplicity-one is not corank-one)**.** The dichotomy in Lemma 3.3 below is sharper than rank bookkeeping, and the following example shows why the distinction matters. Let $m \ge 2$ and

$$\widehat{A} = \begin{pmatrix} x_1 & x_2 \\ 0 & x_1 \end{pmatrix} \oplus x_0 I_{m-2}, \qquad F = \det \widehat{A} = x_0^{m-2} x_1^2.$$

Along the component $S = \{x_1 = 0\}$ the matrix has corank exactly one at a generic point, with left kernel $u = e_2$ and right kernel $v = e_1$, yet S has multiplicity two in F and every kernel pairing $u^{\mathsf{T}} \widehat{A}_i v = (\widehat{A}_i)_{21}$ vanishes: the component is *tangential*. Generic corank one does not imply

multiplicity one; the differential criterion of Lemma 3.3 is the correct test, and Γ_1 is defined through it.

Lemma 3.3 (Multiplicity dichotomy for determinants). *In the setting of Theorem 1.4, let S be an irreducible component of $V(F)$, $F = \det \widehat{A}$. The following are equivalent:*

(a) S has multiplicity one in F;
(b) dF does not vanish identically on S;
(c) at a generic point of S the matrix $\widehat{A}$ has corank exactly one, and *the kernel pairing $w \mapsto u^{\mathsf{T}}\widehat{A}(w)\,v$ (with u, v spanning the left and right kernels) is not identically zero.*

In the symmetric case (c) reads: corank one and $w \mapsto u^{\mathsf{T}}\widehat{A}(w)\,u \not\equiv 0$.

What the dichotomy is testing

Corank one is necessary because a nonzero adjugate is needed to obtain a nonzero derivative of the determinant. It is not sufficient: the affine linear subspace of matrices may be tangent to the determinant hypersurface. The kernel pairing in part (c) tests whether the parameter directions cross the determinant hypersurface transversely rather than merely moving inside its tangent space.

Proof. (a)⇔(b) is the computation in Lemma 3.2. For (b)⇔(c): since $\widehat{A}$ is linear in $(x_0, \dots, x_n)$, the directional derivative of F at x in direction w is, by Jacobi's formula,

$$\partial_w F(x) = \operatorname{tr}\big(\operatorname{adj}\widehat{A}(x) \cdot \widehat{A}(w)\big). \tag{2}$$

If $\widehat{A}$ has corank ≥ 2 at a generic point of S, then $\operatorname{adj}\widehat{A} \equiv 0$ on S and $dF \equiv 0$ on S by (2). If the corank is one at a generic point x, then $\operatorname{adj}\widehat{A}(x)$ is a nonzero rank-one matrix; its columns lie in $\ker \widehat{A}(x)$ and its rows in the left kernel (from $\widehat{A} \cdot \operatorname{adj}\widehat{A} = \operatorname{adj}\widehat{A} \cdot \widehat{A} = F \cdot I = 0$ on S), so $\operatorname{adj}\widehat{A}(x) = \alpha(x)\, v(x)u(x)^{\mathsf{T}}$ with $\alpha(x) \neq 0$, and (2) becomes $\partial_w F(x) = \alpha(x)\, u(x)^{\mathsf{T}}\widehat{A}(w)\, v(x)$. Thus $dF|_S \equiv 0$ if and only if the pairing vanishes identically. In the symmetric case $\operatorname{adj}\widehat{A}(x)$ is symmetric of rank one with image equal to the kernel line, so $\operatorname{adj}\widehat{A}(x) = \beta(x)\, u(x)u(x)^{\mathsf{T}}$, $\beta(x) \neq 0$, and the same computation applies. □

3.2 Border complexity and closures

Lemma 3.4 (Closure agreement). *For a constructible subset Y of an affine variety over $\mathbb{C}$, the Zariski and Euclidean closures of Y coincide.*

Proof. Standard [11, I.10]: Y contains a subset that is open and dense, in both topologies, in the Zariski closure $\overline{Y}^{\mathrm{Zar}}$, and Zariski-closed sets are Euclidean-closed. □

3.3 Transversality and positivity inputs

The following two lemmas are the only intersection-theoretic inputs to Theorem 1.4; each is stated with the hypotheses under which it will be invoked.

Lemma 3.5 (Kleiman transversality, char. 0). *Let $P = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_r}$, let $V_1, \dots, V_s \subset P$ be irreducible subvarieties, and let $V_l^\circ \subseteq V_l$ be dense opens. Let $\Phi \subset P$ be a product of linear subspaces, of codimension equal to $\dim V_l$ for all l. For a generic translate $g\Phi$ under the transitive action of $G = \mathrm{PGL}_{a_1+1} \times \cdots \times \mathrm{PGL}_{a_r+1}$: each intersection $g\Phi \cap V_l$ is finite, reduced (transverse at smooth points of V_l), contained in V_l°, of cardinality $\deg\big([V_l] \cdot [\Phi]\big)$; and the intersections for distinct l are pairwise disjoint. In particular, multidegrees of effective cycles are nonnegative and are computed by counting generic flag intersections componentwise.*

How this lemma is used

The input cycle may have singular points and several components. A generic product of linear spaces misses all prescribed lower-dimensional bad sets, meets each relevant component transversely in its smooth open part, and makes the number of points equal to the corresponding intersection number. No genericity assumption is placed on the cycle itself.

Proof. Kleiman's theorem [6] applied to each V_l (transitive group action, characteristic zero for generic reducedness of the intersection); containment in V_l° and pairwise disjointness are the same statement applied to the lower-dimensional closed sets $V_l \setminus V_l^\circ$ and $V_l \cap V_{l'}$, which a generic translate of Φ misses for dimension reasons. □

Lemma 3.6 (Bézout bound for isolated zeros). *Let $P = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_r}$, let $E = \bigoplus_{j=1}^{D} \mathcal{O}_P(d_j)$ be a direct sum of line bundles with componentwise nonnegative multidegrees d_j, where $D = \dim P$, and let $s \in H^0(P, E)$ be a section with zero scheme $Z = Z(s)$. Then*

$$\sum_{\substack{p \in Z \\ p \text{ isolated in } Z}} \dim_{\mathbb{C}} \mathcal{O}_{Z,p} \;\le\; \int_P c_D(E) \;=\; \Big[\, \textstyle\prod_l H_l^{a_l} \,\Big]\; \prod_{j=1}^{D} \Big(\sum_l d_{j,l} H_l \Big),$$

where H_l are the hyperplane classes and the bracket extracts the coefficient in

$$\mathbb{Z}[H_1, \ldots, H_r] \,/\, (H_1^{a_1+1}, \ldots, H_r^{a_r+1}).$$

Each isolated point contributes its local multiplicity $\dim_{\mathbb{C}} \mathcal{O}_{Z,p} \ge 1$.

Why excess components are allowed

The zero scheme $Z(s)$ need not be finite. It may contain curves or larger components caused by singular or repeated factors. The lemma does not try to remove them. It says only that the sum of the local multiplicities of the isolated points is at most the total top-Chern number, because every excess component contributes a nonnegative localized class. This one-sided estimate is precisely what the determinant proof needs.

Proof. The section s has a localized top Chern class $\mathbf{Z}(s) \in A_0(Z)$ whose image in $A_0(P)$ is $c_D(E) \cap [P]$, of degree $\int_P c_D(E)$ [3, §14.1]. The class decomposes over the connected components of Z. At an isolated point p the local contribution is the intersection multiplicity $\dim_{\mathbb{C}} \mathcal{O}_{Z,p} \ge 1$ [3, §7.1, §14.1]. On every other connected component the contribution is nonnegative, because E is globally generated (each $\mathcal{O}_P(d_j)$ with $d_j \ge 0$ is) and the localized class of a section of a globally generated bundle is represented by a nonnegative cycle [3, §12.2]. Summing gives the bound; the coefficient formula is the Chow ring of a product of projective spaces. (For this statement in the language of multihomogeneous Bézout numbers see also [15, Ch. 8].) □

3.4 Dual degree of a smooth hypersurface

Lemma 3.7 (Classical dual-degree formula). *Let $Z \subset \mathbb{P}^N$ be a smooth hypersurface of degree $e \ge 2$ over $\mathbb{C}$. Then the dual variety $Z^\vee \subset (\mathbb{P}^N)^\vee$ is a hypersurface, the conormal map $\mathrm{Con}(Z) \to Z^\vee$ is birational (reflexivity in characteristic zero), and*

$$\deg Z^\vee \;=\; e(e-1)^{N-1}.$$

Equivalently, the top multidegree of $\mathrm{Con}(Z)$ equals $e(e-1)^{N-1}$.

Proof. Let q be a homogeneous equation of degree e for Z. Smoothness means that the partial derivatives of q do not vanish simultaneously on Z, so they define the Gauss morphism

$$\gamma : Z \longrightarrow (\mathbb{P}^N)^\vee, \qquad x \longmapsto [\partial_0 q(x) : \cdots : \partial_N q(x)].$$

Each coordinate of γ has degree $e-1$; therefore

$$\gamma^* O_{(\mathbb{P}^N)^\vee}(1) = O_Z(e-1).$$

A smooth hypersurface of degree at least two is not dual-defective, so the image $Z^\vee$ is a hypersurface. Reflexivity in characteristic zero implies that the conormal projection to $Z^\vee$ is generically one-to-one; hence the Gauss morphism has generic degree one onto its image [4, Ch. 1], [16, §7]. Intersecting $Z^\vee$ with $N-1$ generic hyperplanes and pulling those hyperplanes back by γ gives

$$\deg Z^\vee = \int_Z c_1\big(O_Z(e-1)\big)^{N-1} = (e-1)^{N-1} \int_Z H^{N-1}.$$

Finally, $\int_Z H^{N-1} = \deg Z = e$, because Z is a degree-e hypersurface. Thus $\deg Z^\vee = e(e-1)^{N-1}$. This is the standard Gauss-map computation of the dual degree [12]. □

4 Bounding the Gauss graph of an arbitrary determinant

What this section contributes

The border argument later treats Theorem 1.4 as a black box. It says that every size-m affine-linear determinant has limited conormal complexity, even when the determinant is singular, reducible, or non-reduced. Readers focused on the specialization argument may read the sketch below and then continue at Section 5.

Proof idea before the details. A generic flag turns $\delta_{n-2}(\Gamma_1(F))$ into a finite set of smooth Gauss-graph points. At each such point the matrix has corank one, so the point lifts uniquely to its left and right kernel lines. We encode these lifted points by a square multihomogeneous system in (x, u, v). A Schur-complement normal form shows that every desired lift is an isolated reduced solution; multihomogeneous Bézout then bounds their number. In the symmetric model the two kernel lines coincide, which removes one projective factor and improves the constant.

4.1 The local calculation that drives the proof

The technically longest part of the argument is a familiar Schur-complement calculation. Suppose that, after constant row and column changes, a matrix near a corank-one point has block form

$$\widehat{A}(x) = \begin{pmatrix} B(x) & c(x) \\ r(x) & s(x) \end{pmatrix}, \qquad \det B(x^*) \neq 0.$$

Then B remains invertible in the local ring at x^*, and

$$\det \widehat{A} = (\det B)g, \qquad g = s - rB^{-1}c.$$

On the hypersurface germ $g = 0$, the projective kernel lines are represented by

$$v(x) = \begin{pmatrix} -B^{-1}c \\ 1 \end{pmatrix}, \qquad u(x)^{\mathsf{T}} = \big(-rB^{-1}, 1\big).$$

Thus the kernel equations do not create a mysterious new local variety: they solve uniquely for u and v as functions of x, while their last remaining condition is exactly the hypersurface equation $g = 0$. Step 5 below proves this statement scheme-theoretically and then shows that the flag equations cut the resulting graph transversely.

For $m = 2$, B is a single invertible scalar function b, and the formulas become

$$\det \begin{pmatrix} b & c \\ r & s \end{pmatrix} = b\left(s - \frac{rc}{b}\right), \quad v = \left(-\frac{c}{b}, 1\right), \quad u^{\mathsf{T}} = \left(-\frac{r}{b}, 1\right).$$

The general calculation is the same, with matrix inverses replacing scalar division.

4.2 Variable and equation bookkeeping in the ordinary model

Block	Number	Multidegree	Purpose
Point $x \in \mathbb{P}^n$	dimension n	H	Location on the hypersurface
Left kernel $[u] \in \mathbb{P}^{m-1}$	dimension $m-1$	U	Encodes the row-kernel line
Right kernel $[v] \in \mathbb{P}^{m-1}$	dimension $m-1$	V	Encodes the column-kernel line
$u^{\mathsf{T}}\widehat{A}(x) = 0$	m equations	$H+U$	Full left-kernel condition
$\Lambda\widehat{A}(x)v = 0$	$m-1$ equations	$H+V$	Locally complete right-kernel condition
Dual flag equations	$n-2$ equations	$U+V$	Conditions on the tangent hyperplane
Primal flag equation	1 equation	H	One condition on x

The ambient dimension is $n+2m-2$, and the total number of equations is also $n+2m-2$. Squareness alone does not prove finiteness; Step 5 verifies that the particular lifts we care about are indeed reduced isolated solutions.

Throughout, $A, \widehat{A}, F$ are as in Theorem 1.4, $n \geq 2$, and $F = \det \widehat{A} \not\equiv 0$. If $\Gamma_1(F)$ is zero there is nothing to prove. Otherwise write $\Gamma_1(F) = \sum_{j\in J} \mathrm{Con}(S_j)$ as in Lemma 3.2.

4.3 Proof of Theorem 1.4(i)

Step 1: count points.

Roadmap: turn the multidegree into a finite list

Each conormal component has dimension $n-1$. Impose one generic primal hyperplane and $n-2$ generic dual hyperplanes, for a total codimension $n-1$. Kleiman transversality makes the intersection finite, reduced, and contained in the smooth open Gauss graph. The output is exactly $\delta_{n-2}(\Gamma_1(F))$ distinct points together with the linear independence of the corresponding flag differentials. That differential independence is saved for Step 5.

For each $j \in J$ let $S_j^{\mathrm{good}} \subseteq S_j$ be the dense open subset of points x such that x is a smooth point of $V(F)_{\mathrm{red}}$ lying on S_j only, and $dF(x) \neq 0$; this is dense because S_j has multiplicity one (Lemma 3.3(b)) and F restricted to a neighborhood of a generic point of S_j is a unit times a reduced equation of S_j. Over S_j^{good} the conormal variety is the graph $x \mapsto (x, [dF(x)])$; let $\mathrm{Con}(S_j)^\circ$ be this graph, dense open in $\mathrm{Con}(S_j)$, isomorphic to S_j^{good} via the first projection.

Apply Lemma 3.5 on $\mathbb{P}^n \times (\mathbb{P}^n)^\vee$ to the varieties $\{\mathrm{Con}(S_j)\}_{j\in J}$, the opens $\mathrm{Con}(S_j)^\circ$, and the flag $\Phi = M \times (L_1 \cap \cdots \cap L_{n-2})$ with M a hyperplane in $\mathbb{P}^n$ and L_t hyperplanes in $(\mathbb{P}^n)^\vee$: for a generic such flag, the intersection consists of exactly $\sum_j \delta_{n-2}(\mathrm{Con}(S_j)) = \delta_{n-2}(\Gamma_1(F))$ distinct *count points*

$$(x^*, \xi^*), \qquad x^* \in S_j^{\mathrm{good}}, \quad \xi^* = [dF(x^*)],$$

each a transverse, reduced intersection point of the flag with the graph $\mathrm{Con}(S_j)^\circ$. Transversality at (x^*, ξ^*), transported through the isomorphism $S_j^{\mathrm{good}} \cong \mathrm{Con}(S_j)^\circ$, says precisely: writing ℓ_t for the linear forms cutting $L_1 \cap \cdots \cap L_{n-2}$ and M for the linear form cutting the hyperplane, the $n-1$

functions

$$M(x), \quad \ell_1\big(dF(x)\big), \ \ldots, \ \ell_{n-2}\big(dF(x)\big) \tag{3}$$

vanish at x^* and have linearly independent differentials on $T_{x^*}S_j$. This is the form in which transversality will be used in Step 5.

Step 2: the kernel lift.

Roadmap: replace each tangent point by kernel data

At a count point, $F(x^*) = 0$ but $dF(x^*) \neq 0$. Jacobi's formula therefore forces the adjugate to be nonzero, so the matrix has corank exactly one. Consequently its projective left and right kernels are unique. The rank-one adjugate identity then expresses every coordinate of the tangent covector as $u^{\mathsf{T}}\widehat{A}_i v$. The output is one unique lift $(x^*, [u^*], [v^*])$ for each Gauss point.

Fix a count point (x^*, ξ^*). Since $dF(x^*) \neq 0$, formula (2) forces $\operatorname{adj}\widehat{A}(x^*) \neq 0$, hence $\operatorname{rank}\widehat{A}(x^*) \geq m-1$; since $F(x^*) = 0$ the rank is exactly $m-1$. The left and right kernels are lines, spanned by u^*, v^*, and as in the proof of Lemma 3.3,

$$\operatorname{adj}\widehat{A}(x^*) = \alpha\, v^*(u^*)^{\mathsf{T}}, \quad \alpha \neq 0, \qquad \partial_i F(x^*) = \alpha\,(u^*)^{\mathsf{T}}\widehat{A}_i\, v^* \ \ (0 \leq i \leq n), \tag{4}$$

so $\xi^* = \big[(u^*)^{\mathsf{T}}\widehat{A}_0 v^* \ : \ \cdots \ : \ (u^*)^{\mathsf{T}}\widehat{A}_n v^*\big]$. Each count point therefore lifts to the unique point $P^* = (x^*, [u^*], [v^*]) \in \mathbb{P}^n \times \mathbb{P}^{m-1} \times \mathbb{P}^{m-1}$.

Step 3: choice of the reduction matrix Λ.

Roadmap: remove exactly one redundant kernel equation

The two full kernel systems contain $2m$ scalar equations, but the corank-one kernel incidence has codimension $2m-1$. The relation $u^{\mathsf{T}}(\widehat{A}v) = (u^{\mathsf{T}}\widehat{A})v$ explains why the two blocks cannot be independent. We retain all left equations and replace the m right equations by $m-1$ generic linear combinations. The map Λ is chosen only after the finite list of count points is known, so it can be made injective on every relevant image hyperplane simultaneously. Step 5 proves that this reduced block defines the same local kernel line at each lift.

The incidence conditions at a lift are $u^{\mathsf{T}}\widehat{A}(x) = 0$ (m equations) and $\widehat{A}(x)v = 0$ (m equations), but the incidence variety over the corank-one locus has codimension $2m-1$, not $2m$, in $\mathbb{P}^n \times \mathbb{P}^{m-1} \times \mathbb{P}^{m-1}$: one equation is locally redundant. We remove the redundancy on the *right* side. Choose a linear map $\Lambda : \mathbb{C}^m \to \mathbb{C}^{m-1}$, generic subject to the finitely many open conditions

$$\ker\Lambda \cap \operatorname{im}\widehat{A}(x^*) = 0 \qquad \text{for every count point } x^*, \tag{5}$$

each of which excludes a proper closed subset of Λ-space because $\ker\Lambda$ is a line and $\operatorname{im}\widehat{A}(x^*)$ is a hyperplane of $\mathbb{C}^m$. We also require, generically, that no row of $\Lambda\widehat{A}$ vanishes identically as a form (the left annihilators $\{\lambda : \lambda^{\mathsf{T}}\widehat{A} \equiv 0\}$ form a proper linear subspace since $\widehat{A} \not\equiv 0$). The scheme-theoretic content of the reduction — that near each lift the $m-1$ equations $\Lambda\widehat{A}v = 0$ generate the same ideal as the full $\widehat{A}v = 0$ — is established by the explicit elimination in Step 5(ii), where condition (5) resurfaces as the invertibility of a concrete $(m-1)\times(m-1)$ matrix.

Remark 4.1 (Why a row may not simply be deleted). Replacing the m equations $\widehat{A}v = 0$ by $m-1$ of them — deleting a row instead of taking $m-1$ generic combinations — is unsound: deleting row k is the choice $\Lambda =$ (projection killing e_k), for which the matrix M_v of Step 5(ii) is a fixed $(m-1)\times(m-1)$ submatrix of $\widehat{A}$'s column block, which can be singular at special x^* even where $\widehat{A}$ has corank one; the solution set then fattens to a positive-dimensional kernel through the lift and

Step 5 fails. The generic-Λ reduction keeps the solution set equal to the kernel line near every count point. The same caution applies on the left: the left system is kept *whole* (m equations), because it is what excludes solutions lying over the locus $\det \widehat{A}(x) \neq 0$ in Step 5(i).

Step 4: the square multihomogeneous system.

Roadmap: write one global system to which Bézout applies

Combine the left-kernel equations, the reduced right-kernel equations, the $n-2$ dual flag equations, and the one primal flag equation. Their multidegrees are read separately in the x, u, and v variables. The top Chern class of the resulting direct sum of line bundles is the coefficient $B(m,n)$. At this stage Bézout gives only a total intersection budget; it remains to show that every desired lift consumes at least one unit of that budget as an isolated point.

On $Y = \mathbb{P}^n \times \mathbb{P}^{m-1}_{[u]} \times \mathbb{P}^{m-1}_{[v]}$, with hyperplane classes H, U, V and $\dim Y = n+2m-2$, consider the system

$$\begin{array}{lll} u^{\mathsf{T}}\widehat{A}(x) = 0 & m \text{ equations} & \text{class } H+U, \\ \Lambda\,\widehat{A}(x)\,v = 0 & m-1 \text{ equations} & \text{class } H+V, \\ \ell_t\big(u^{\mathsf{T}}\widehat{A}_0 v, \ldots, u^{\mathsf{T}}\widehat{A}_n v\big) = 0, \quad 1 \le t \le n-2 & n-2 \text{ equations} & \text{class } U+V, \\ M(x) = 0 & 1 \text{ equation} & \text{class } H, \end{array} \tag{6}$$

where ℓ_t, M are the flag forms of Step 1. The count is $m+(m-1)+(n-2)+1 = n+2m-2 = \dim Y$: the system is square. The class assignments: each entry of $u^{\mathsf{T}}\widehat{A}$ is bilinear in (x,u) and each entry of $\Lambda\widehat{A}v$ is bilinear in (x,v); each polar equation is $\ell_t(\ldots) = u^{\mathsf{T}}\big(\sum_i \lambda_{t,i}\widehat{A}_i\big)v$, bilinear in (u,v) and independent of x — a genuine nonzero section of $\mathcal{O}(0,1,1)$ for generic ℓ_t, since $\sum_i \lambda_i \widehat{A}_i = 0$ defines a proper linear subspace of covectors ($\widehat{A} \not\equiv 0$), which the generic flag avoids. Likewise each entry of $u^{\mathsf{T}}\widehat{A}$ is a nonzero section (a zero entry would be a zero column of $\widehat{A}$, forcing $F \equiv 0$), and the rows of $\Lambda\widehat{A}$ were arranged nonzero in Step 3. All multidegrees are componentwise nonnegative, so Lemma 3.6 applies with

$$\int_Y c_{\mathrm{top}} = \big[H^n U^{m-1} V^{m-1}\big]\; H\,(H+U)^m (H+V)^{m-1} (U+V)^{n-2} \;=\; B(m,n),$$

the equality of the two displayed brackets being the symmetry $u \leftrightarrow v$ of the extraction (the closed form is computed in Step 6).

Step 5: the lifts are reduced isolated points of the solution scheme.

Technical heart of the determinant-side proof

A square system can still have positive-dimensional components, and such components do occur over repeated or higher-corank pieces of the determinant. The task here is narrower: prove that each lift selected in Step 1 is locally a single reduced point of the full system. The proof first shows that the kernel block is a graph over the smooth hypersurface germ, and then invokes the transversality already obtained in Step 1 to cut that graph to one point.

Fix a count point with lift $P^* = (x^*, [u^*], [v^*])$.

(i) No solutions off the hypersurface. If $\det \widehat{A}(x) \neq 0$ then $u^{\mathsf{T}}\widehat{A}(x) = 0$ forces $u = 0$, which is not a point of $\mathbb{P}^{m-1}$. This is why the left system is kept whole.

(ii) Local normal form: the bilinear block is scheme-theoretically a graph over the hypersurface germ. After constant row and column operations $\widehat{A} \mapsto P\widehat{A}Q$ — which multiply F by the nonzero constant $\det P \det Q$, transform $u \mapsto P^{-\mathsf{T}}u$, $v \mapsto Q^{-1}v$, $\Lambda \mapsto \Lambda P^{-1}$ equivariantly, and leave every

pairing $u^{\mathsf{T}}\widehat{A}_i v$ and condition (5) intact — we may assume the leading $(m-1)\times(m-1)$ block of $\widehat{A}(x^*)$ is invertible (some $(m-1)$-minor is nonzero at a rank-$(m-1)$ matrix). Work in an affine chart of $\mathbb{P}^n$ at x^* and write

$$\widehat{A} = \begin{pmatrix} B & c \\ r & s \end{pmatrix}, \qquad \det B \in \mathcal{O}_{x^*}^{\times},$$

with B of size $m-1$ and all blocks affine-linear in the chart coordinates. Then $u_m^* \neq 0$ and $v_m^* \neq 0$: a kernel vector with vanishing last coordinate is annihilated by the invertible B and dies. So we may work in the charts $u_m = 1$, $v_m = 1$, writing $u^{\mathsf{T}} = (u'^{\mathsf{T}}, 1)$, $v = (v', 1)^{\mathsf{T}}$. Set

$$g \;:=\; s - rB^{-1}c \;\in \mathcal{O}_{x^*}, \qquad z \;:=\; v' + B^{-1}c,$$

the Schur complement and the recentred right-kernel coordinate, and recall the Schur identity $\det \widehat{A} = (\det B)\, g$, so that $(F) = (g)$ in $\mathcal{O}_{x^*}$. In the local ring $\mathcal{O}_{P^*}$ (variables: chart coordinates of x, and u', v'), the ideal I of the bilinear block $\{u^{\mathsf{T}}\widehat{A} = 0,\ \Lambda\widehat{A}v = 0\}$ is computed by elimination.

Left block. The first $m-1$ left equations are $u'^{\mathsf{T}}B + r = 0$; since B is invertible over $\mathcal{O}_{x^*}$ they generate the same ideal as the entries of $u'^{\mathsf{T}} + rB^{-1}$. Modulo these, the last left equation becomes $u'^{\mathsf{T}}c + s \;\equiv\; (-rB^{-1})c + s \;=\; g$.

Right block. Writing $v' = z - B^{-1}c$,

$$\widehat{A}\,v = \begin{pmatrix} Bv' + c \\ rv' + s \end{pmatrix} = \begin{pmatrix} Bz \\ rz + g \end{pmatrix}, \qquad \text{hence} \qquad \Lambda\widehat{A}\,v \;=\; M_v(x)\, z \;+\; g\,\lambda,$$

an exact identity, where $\Lambda = [\Lambda_{\mathrm{top}} \mid \lambda]$ with λ the last column of Λ, and

$$M_v(x) \;:=\; \Lambda\, C(x), \qquad C(x) := \begin{pmatrix} B \\ r \end{pmatrix} = \text{the first } m-1 \text{ columns of } \widehat{A}.$$

At x^*, $C(x^*)$ has rank $m-1$ (its top block is $B(x^*)$) and its column space is therefore all of $\operatorname{im}\widehat{A}(x^*)$; so condition (5) says *exactly* that Λ is injective on the column space of $C(x^*)$, i.e. that $M_v(x^*)$ is invertible. Hence M_v is invertible over $\mathcal{O}_{x^*}$, and modulo the generators already obtained, $\Lambda\widehat{A}v \equiv M_v z$ generates the same ideal as the entries of z. Altogether

$$I \;=\; \left(u'^{\mathsf{T}} + rB^{-1},\ \ v' + B^{-1}c,\ \ g\right) : \tag{7}$$

This equality of ideals is stronger than equality of point sets. The first $m-1$ generators solve uniquely for the affine coordinates u', the next $m-1$ solve uniquely for v', and the final generator is the single equation $g = 0$ for the determinant hypersurface. In particular there is no hidden nilpotent thickening in the kernel directions: the local quotient ring is canonically the local ring of $V(g)$.

Consequently, the bilinear block is, scheme-theoretically, the graph

$$x \;\longmapsto\; \big(x,\ [u(x)],\ [v(x)]\big), \qquad u(x)^{\mathsf{T}} = (-rB^{-1},\ 1), \quad v(x) = (-B^{-1}c,\ 1)^{\mathsf{T}},$$

over the germ $V(g)$. Since $(F) = (g)$ and $dF(x^*) \neq 0$, that germ is the smooth reduced hypersurface germ (S_j, x^*), and the graph is a smooth germ of dimension $n-1$ through P^*.

(iii) The conormal identity along the graph, and transversality. Along (S_j, x^*) the explicit sections satisfy $\widehat{A}(x)v(x) = (Bz, rz+g)^{\mathsf{T}}\big|_{z=0,\,g=0} = 0$ and $u(x)^{\mathsf{T}}\widehat{A}(x) = (0,\ g)\big|_{g=0} = 0$: they span the kernels. At each such point $\widehat{A}$ has rank $m-1$, so $\operatorname{adj}\widehat{A}(x) = \alpha(x)\, v(x)u(x)^{\mathsf{T}}$; evaluating the (m,m) entries pins the scalar, since $(\operatorname{adj}\widehat{A})_{mm} = \det B$ and $(vu^{\mathsf{T}})_{mm} = 1$, giving $\alpha = \det B$, a unit. Jacobi's formula (2) then yields, on (S_j, x^*) and for every i,

$$\partial_i F(x) \;=\; (\det B)(x) \cdot u(x)^{\mathsf{T}}\widehat{A}_i\, v(x).$$

Consequently each polar equation of (6) restricts on the graph to $(\det B)^{-1}\,\ell_t(dF(x))$, a unit multiple of the corresponding flag function (3); both vanish at x^*, so their differentials at x^* agree up to the unit value $(\det B)(x^*)^{-1}$. The remaining equation $M(x)$ restricts to itself. By Step 1, the $n-1$ flag functions (3) have linearly independent differentials on $T_{x^*}S_j$. Therefore the full system (6) cuts, near P^*, the smooth $(n-1)$-dimensional germ (7) by $n-1$ functions with independent differentials: a single reduced point. P^* is a reduced isolated point of the solution scheme, and distinct count points have distinct x^*, hence distinct lifts.

The role of the count-point condition is visible in the normal form: the last left equation contributes the hypersurface equation g, and the polar equations contribute transverse slices precisely because $dF(x^*) \neq 0$. On a tangential component (Example 3.2) every lifted polar equation would restrict to zero along the graph and no isolated point would form; the dichotomy of Lemma 3.3 is what the construction runs on.

Step 6: conclusion and the closed form.

Roadmap: charge the desired points to the Bézout budget

Step 5 gives one reduced isolated solution for every multidegree count point. Lemma 3.6 bounds the sum of multiplicities of all isolated solutions by the top-Chern coefficient, even if excess components are also present. The final task is purely combinatorial: expand the coefficient and match the exponents of H, U, V.

By Lemma 3.6, the number of isolated solutions of (6), counted with multiplicity ≥ 1, is at most $B(m,n)$; by Step 5 the $\delta_{n-2}(\Gamma_1(F))$ lifts are among them, each reduced. Hence $\delta_{n-2}(\Gamma_1(F)) \leq B(m,n)$. For the closed form, extract the bracket. Choose $x^i u^{m-i}$ from $(x+u)^m$, choose $x^j v^{m-1-j}$ from $(x+v)^{m-1}$, and choose $u^k v^{n-2-k}$ from $(u+v)^{n-2}$. Including the leading factor x, the target monomial $x^n u^{m-1} v^{m-1}$ imposes

variable	exponent equation	consequence
x	$1+i+j=n$	$j=n-1-i$
u	$(m-i)+k=m-1$	$k=i-1$
v	$(m-1-j)+(n-2-k)=m-1$	automatic after the first two

Thus the free index is i, with $1 \leq i \leq n-1$, and

$$B(m,n) = \sum_{i=1}^{n-1} \binom{m}{i}\binom{m-1}{n-1-i}\binom{n-2}{i-1},$$

the limits enforced by $k \geq 0$ and $j \geq 0$. □

Proof check: ordinary determinant bound

The proof did not assume that $V(F)$ is smooth, irreducible, reduced, or generically corank one on every component. Generic flags were used only to select count points on the multiplicity-one components. Higher-corank and tangential components may create excess solution sets, but Lemma 3.6 allows them. The generic reduction matrix Λ was chosen after the finite count set, and Step 5 verified its local scheme-theoretic validity at every desired lift.

4.4 Proof of Theorem 1.4(ii)

The symmetric proof is given in full; it runs on $Y = \mathbb{P}^n \times \mathbb{P}^{m-1}_{[u]}$, of dimension $n+m-1$.

	Ordinary model	Symmetric model
Kernel data	Separate left and right lines $[u]$, $[v]$	One kernel line $[u]$
Ambient space	$\mathbb{P}^n \times \mathbb{P}^{m-1} \times \mathbb{P}^{m-1}$	$\mathbb{P}^n \times \mathbb{P}^{m-1}$
Tangent coordinates	$u^{\mathsf{T}}\widehat{A}_i v$	$u^{\mathsf{T}}\widehat{A}_i u$
Dual flag class	$U + V$	$2U$
Kernel reduction	One generic Λ is needed	No reduction is needed
Final coefficient	$B(m, n)$	$2^{n-2}\binom{m}{n-1}$

Step 1: count points.

What is unchanged

The multidegree and the multiplicity-one Gauss graph depend only on the polynomial F, not on whether its determinantal representation is symmetric. The same generic flag therefore produces the same finite set of transverse count points.

Step 1 of Section 4.3 concerns only the cycle $\Gamma_1(F)$ and not the representation; it applies verbatim and furnishes $\delta_{n-2}(\Gamma_1(F))$ count points (x^*, ξ^*) with $x^* \in S_j^{\mathrm{good}}$, $\xi^* = [dF(x^*)]$, together with the transversality statement for the flag functions (3) on $T_{x^*}S_j$.

Step 2: the kernel lift.

Why one kernel line suffices

For a symmetric corank-one matrix, the left and right kernels coincide. Its rank-one adjugate is symmetric and hence a scalar multiple of uu^{T}. This replaces bilinear tangent coordinates by quadratic coordinates and removes one projective factor from the incidence.

At a count point, $dF(x^*) \neq 0$ and (2) force $\operatorname{adj}\widehat{A}(x^*) \neq 0$, hence $\operatorname{rank}\widehat{A}(x^*) = m - 1$ as before. Since $\widehat{A}(x^*)$ is symmetric, its adjugate is a nonzero *symmetric* rank-one matrix whose image is the kernel line, so

$$\operatorname{adj}\widehat{A}(x^*) = \beta\, u^*(u^*)^{\mathsf{T}}, \quad \beta \neq 0, \qquad \partial_i F(x^*) = \beta\,(u^*)^{\mathsf{T}}\widehat{A}_i\, u^* \ \ (0 \le i \le n),$$

and the count point lifts to the unique $P^* = (x^*, [u^*]) \in \mathbb{P}^n \times \mathbb{P}^{m-1}$.

Step 3: squareness without a reduction.

Equation count

The m kernel equations, the $n - 2$ dual flag equations, and the one primal flag equation total $m + n - 1$, exactly the dimension of $\mathbb{P}^n \times \mathbb{P}^{m-1}$. The last kernel equation supplies the local hypersurface equation, so deleting one would be unsound.

The kernel condition $\widehat{A}(x)u = 0$ is m equations, and the system below has $m + (n - 2) + 1 = n + m - 1 = \dim Y$ equations: square as it stands. No analogue of the Λ-reduction is needed — the elimination in Step 5(ii) shows the m kernel equations cut exactly codimension m near the lift, the last of them supplying the hypersurface equation — and none is sound to add: removing an equation enlarges the local solution set to the kernel of an $(m - 1) \times m$ submatrix, which can be positive-dimensional through the lift, exactly as in Remark 4.1.

Step 4: the system and its classes.

Where the factor 2^{n-2} comes from

Each tangent coordinate is quadratic in the single kernel vector u. Therefore each of the $n-2$ dual flag equations has class $2U$, contributing one factor of 2 to the top-Chern coefficient.

The equations are

$$\begin{array}{lll} \widehat{A}(x)\,u = 0 & m \text{ equations} & \text{class } H+U, \\ \ell_t\big(u^{\mathsf T}\widehat{A}_0 u, \dots, u^{\mathsf T}\widehat{A}_n u\big) = 0, \;\; 1 \le t \le n-2 & n-2 \text{ equations} & \text{class } 2U, \\ M(x) = 0 & 1 \text{ equation} & \text{class } H, \end{array} \tag{8}$$

with ℓ_t, M the flag forms of Step 1. Each kernel entry is bilinear in (x,u); each polar equation is $u^{\mathsf T}(\sum_i \lambda_{t,i}\widehat{A}_i)u$, a quadratic form in u alone. Nonzero sections: a vanishing row of $\widehat{A}$ forces $F \equiv 0$; and a flag form with $u^{\mathsf T}(\sum_i \lambda_i\widehat{A}_i)u \equiv 0$ forces $\sum_i \lambda_i\widehat{A}_i = 0$ — a symmetric matrix over $\mathbb{C}$ whose quadratic form vanishes identically is zero in characteristic zero — which is a proper linear condition on λ avoided by the generic flag. All multidegrees are componentwise nonnegative.

Step 5: the lifts are reduced isolated points.

Symmetric local normal form

A congruence change makes the leading block invertible. The first $m-1$ kernel equations solve for u', and the last becomes the Schur complement g. The kernel incidence is therefore a graph over the smooth hypersurface germ, exactly as in the ordinary case but with no separate right-kernel variables.

Fix a count point with lift $P^* = (x^*, [u^*])$.

(i) No solutions off the hypersurface. Where $\det\widehat{A}(x) \neq 0$, $\widehat{A}(x)u = 0$ forces $u = 0$, not a point of $\mathbb{P}^{m-1}$.

(ii) Local normal form. After a constant congruence $\widehat{A} \mapsto P^{\mathsf T}\widehat{A}P$ — which multiplies F by $(\det P)^2 \neq 0$, transforms $u \mapsto P^{-1}u$, preserves symmetry and every quadratic pairing $u^{\mathsf T}\widehat{A}_i u$ — assume the leading $(m-1)\times(m-1)$ block is invertible at x^*, and write, in an affine chart at x^*,

$$\widehat{A} = \begin{pmatrix} B & c \\ c^{\mathsf T} & s \end{pmatrix}, \qquad \det B \in \mathcal{O}_{x^*}^{\times}, \qquad B = B^{\mathsf T}.$$

Then $u^*_m \neq 0$ (a kernel vector with last coordinate zero dies against B); work in the chart $u_m = 1$, $u = (u', 1)^{\mathsf T}$. Set $g := s - c^{\mathsf T}B^{-1}c$ and $z := u' + B^{-1}c$, so

$$\widehat{A}\,u = \begin{pmatrix} Bu' + c \\ c^{\mathsf T}u' + s \end{pmatrix} = \begin{pmatrix} Bz \\ c^{\mathsf T}z + g \end{pmatrix}, \qquad \det\widehat{A} = (\det B)\, g.$$

Since B is invertible over $\mathcal{O}_{x^*}$, the first $m-1$ entries generate the ideal of the entries of z, and modulo these the last entry becomes g. Hence the ideal of the kernel block in $\mathcal{O}_{P^*}$ is exactly

$$\big(\, u' + B^{-1}c, \;\; g \,\big) :$$

the kernel block is, scheme-theoretically, the graph $u = v(x) := (-B^{-1}c,\, 1)^{\mathsf T}$ over the germ $V(g)$, which equals the smooth reduced germ (S_j, x^*) because $(F) = (g)$ and $dF(x^*) \neq 0$; a smooth germ of dimension $n-1$ through P^*.

(iii) Conormal identity and transversality. Along (S_j, x^*) the section $v(x)$ spans the kernel ($\widehat{A}v = (Bz, c^{\mathsf T}z + g)^{\mathsf T}|_{z=0,g=0} = 0$), the rank is $m-1$, and the symmetric adjugate is a scalar multiple of $vv^{\mathsf T}$; the (m,m) cofactor is $\det B$ while $(vv^{\mathsf T})_{mm} = 1$, so $\operatorname{adj}\widehat{A} = (\det B)\, vv^{\mathsf T}$ on (S_j, x^*), and Jacobi's

formula (2) gives

$$\partial_i F(x) \;=\; (\det B)(x)\cdot v(x)^{\mathsf{T}}\widehat{A}_i\, v(x) \qquad \text{on } (S_j, x^*).$$

Each polar equation of (8) therefore restricts on the graph to $(\det B)^{-1}\ell_t(dF(x))$, a unit multiple of the flag function, vanishing at x^*; $M(x)$ restricts to itself; and by Step 1 the $n-1$ flag functions (3) have independent differentials on $T_{x^*}S_j$. The full system (8) thus cuts, near P^*, the smooth $(n-1)$-dimensional graph by $n-1$ functions with independent differentials: a single reduced point. Distinct count points give distinct lifts.

Step 6: conclusion and the coefficient.

Final extraction

The top-Chern class is $H(H+U)^m(2U)^{n-2}$. To obtain H^nU^{m-1}, the factor $(H+U)^m$ must contribute $H^{n-1}U^{m-n+1}$, whose coefficient is $\binom{m}{n-1}$.

Lemma 3.6 with $E = \mathcal{O}(1,1)^{\oplus m}\oplus\mathcal{O}(1,0)\oplus\mathcal{O}(0,2)^{\oplus(n-2)}$ bounds the isolated solutions of (8) by

$$\left[H^nU^{m-1}\right]H\,(H+U)^m(2U)^{n-2} \;=\; 2^{n-2}\left[H^{n-1}U^{m-n+1}\right](H+U)^m \;=\; 2^{n-2}\binom{m}{n-1},$$

and by Step 5 the $\delta_{n-2}(\Gamma_1(F))$ lifts are among them, each reduced. □

Proof check: why the symmetric constant is better

The geometric lower count is unchanged. The improvement comes entirely from the determinant-side incidence: symmetry identifies the two kernel lines, removes one projective factor and $m-1$ kernel equations, and changes each bilinear dual condition into a quadratic condition on one variable. The resulting coefficient has exponential base $2em/(n-1)$ rather than $4em/(n-1)$.

Remark 4.2 (What multiplicity-≥ 2 components do, and why no codimension hypothesis is needed). Components of $V(F)$ of multiplicity ≥ 2 are of two kinds by Lemma 3.3: *corank-*≥ 2 components, over which the kernel fibres of the incidence are positive-dimensional, and *tangential* corank-one components (Example 3.2), over which the kernel incidence is still a graph but every lifted polar equation vanishes identically. Both populate the solution schemes of (6) and (8) with positive-dimensional excess components. Neither is bounded, located, nor assumed away: Lemma 3.6 charges excess components a nonnegative amount and the isolated lifts at most the stated coefficient, which is all that is used. In particular the corank-≥ 2 locus may be a divisor in $V(F)$ — e.g. $\widehat{A} = \operatorname{diag}(x_1,\dots,x_n,x_0,\dots,x_0)$ has corank ≥ 2 along every $\{x_i = x_j = 0\}$ — and no hypothesis excludes this.

Remark 4.3 (Recovery of the exact companions). Let f be homogeneous of degree $d\geq 2$ in $N\geq 3$ variables with $V(f)\subset\mathbb{P}^{N-1}$ *smooth*, and $f=\det M$ with M of size m. Then $F := \det\widehat{M} = x_0^{m-d}f$ on $\mathbb{P}^N$, the cone $V(f)\subset\mathbb{P}^N$ is a multiplicity-one component of $V(F)$ (f is squarefree and $x_0\nmid f$), so $\Gamma_1(F)\supseteq \operatorname{Con}(\operatorname{cone} V(f))$ as cycles, and Theorem 1.4(i) with $n=N$ plus the cone-shift identity (Proposition 9.1, with Lemma 3.7) gives

$$d(d-1)^{N-2} = \deg V(f)^\vee = \delta_{N-2}\big(\operatorname{Con}(\operatorname{cone} V(f))\big) \leq \delta_{N-2}\big(\Gamma_1(F)\big) \leq B(m,N),$$

which is the main inequality of [13] (the coefficient polynomials agree under the reindexing $a=i-1$); the symmetric variant recovers that of [14] the same way. The hypotheses here are weaker — smoothness of $V(\det\widehat{A})$ is nowhere assumed — which is exactly the strengthening

the border transfer requires, since the generic members of a degenerating family come with no smoothness guarantee.

Remark 4.4 (The reduced conormal). For squarefree F, $\Gamma_1(F)$ is the full reduced conormal cycle and Theorem 1.4 bounds it. Whether δ_{n-2} of the reduced conormal cycle of $V(\det \widehat{A})_{\mathrm{red}}$ obeys the same bound *without* squarefreeness is open; the natural attack (perturb $\widehat{A}$ to make the determinant squarefree and specialize back) is itself a degeneration argument of the kind in Sections 7–8, and we record the question rather than entangle the two halves of this paper. Nothing below needs it: the border chain is built on Γ_1 consistently on both sides.

5 From border membership to a one-parameter family

Main message

Membership in the closure $\overline{D_m}$ can be tested along a curve. We therefore replace the abstract border representation by a regular polynomial family G_c with $G_0 = F_n$ and with each generic fibre admitting some exact size-m determinantal representation. No coherent matrix family is needed.

Why a curve is enough

The set D_m sits in the finite-dimensional affine space of polynomials of degree at most m. A point in its closure can be approached inside the closure along an algebraic curve. Working over a curve is the simplest setting in which specialization has a single parameter, local rings at closed points are discrete valuation rings after normalization, and fibre cycles can be compared directly. The construction does *not* claim that one can choose the matrices A_c algebraically in c; only the polynomials G_c form a coherent family.

From here through Section 10, fix $n \geq 3$ and suppose $\overline{\mathrm{dc}}(F_n) \leq m$ (the symmetric case is identical with D_m^{sym} and is taken up in Section 10).

Lemma 5.1 (One-parameter normal form, fiberwise representations). *There exist a smooth irreducible affine curve C over $\mathbb{C}$, a closed point $0 \in C$, and a family $G \in \mathcal{O}(C)[x_1, \ldots, x_n]_{\leq m}$ with regular coefficient functions, such that:*

(a) $G_0 = F_n$;

(b) there is a dense open $C^\circ \subseteq C \setminus \{0\}$ such that for every closed point $c \in C^\circ$ there exists $A_c \in \mathrm{Mat}_m(\mathbb{C}[x]_{\leq 1})$ with $\det A_c = G_c$ exactly.

Roadmap: from a closure point to fibrewise exact representations

There are two cases. If F_n already has an exact size-m representation, use a constant family. Otherwise, curve selection gives an irreducible curve through F_n whose general point lies in D_m. Normalize the curve so that the parameter space is smooth. Pulling back the coordinate functions of polynomial space gives the regular coefficients of G. Membership of a fibre in D_m then supplies an exact representation for that fibre by definition.

Proof. If $F_n \in D_m$, take $C = \mathbb{A}^1$, G the constant family, and A_c a fixed representation. Otherwise $F_n \in \overline{D_m} \setminus D_m$. By algebraic curve selection through a constructible set [11, I.10] (in the border-complexity setting, [2, §20.6]), since D_m is irreducible constructible in the affine space $\mathbb{C}[x]_{\leq m}$ and $F_n \in \overline{D_m}$, there is an irreducible affine curve $C' \subseteq \overline{D_m}$ with $F_n \in C'$ and $C' \cap D_m$ dense in C'. Let

$\nu : C \to C'$ be the normalization restricted to an affine neighborhood of a point $0 \in \nu^{-1}(F_n)$; C is a smooth irreducible affine curve. The composite $C \to C' \hookrightarrow \mathbb{C}[x]_{\le m}$ is a morphism to affine space, i.e. a tuple of regular functions on C: these are the coefficients of G, and $G_0 = F_n$. The set $\nu^{-1}(C' \cap D_m)$ is dense constructible in the curve C, hence contains a dense open, which we shrink to a C° avoiding 0. For $c \in C^\circ$, the statement $G_c \in D_m$ *is*, by definition of D_m, the existence of an exact affine-linear representation A_c over $\mathbb{C}$. □

Remark 5.1 (What the normal form does *not* provide). No matrix family over $\mathcal{O}(C^\circ)$, $\mathbb{C}[t, t^{-1}]$, or $\mathbb{C}((t))$ is constructed, because nothing downstream consumes one: Sections 6–10 use only the regular polynomial family G and the fiberwise representations A_c, with no compatibility imposed across fibres. This removes any field-of-definition or Laurent-truncation step from the normal form. The single base change in the proof (the normalization ν) alters neither the hypothesis — $F_n \in \overline{D_m}$ concerns the fixed set D_m — nor the parameter m.

Conventions for the family. Several later statements hold after deleting finitely many exceptional parameter values. Instead of introducing a new open set each time, we keep a single finite exclusion set Σ and enlarge it when necessary. Removing finitely many points never destroys the existence of parameters approaching 0 along a punctured analytic disk.

Fix once and for all a finite subset $\Sigma \subset C$ of *excluded points*, enlarged finitely many times in Lemmas 5.2, 7.2, and 7.3 below, and set $U := C^\circ \setminus \Sigma$, always cofinite in C° and in particular infinite. Statements "for $c \in U$" hold for every closed point of U. Analytic limits "$c \to 0$" are taken along an analytic disk in C^{an} centered at 0; since Σ is finite, every punctured analytic neighborhood of 0 meets U in a set accumulating at 0.

Lemma 5.2. *After enlarging Σ: $G_c \neq 0$ for every $c \in U$.*

Proof. The coefficients of G are regular on C and not all identically zero ($G_0 = F_n \neq 0$); a nonzero regular function on a curve vanishes at finitely many points. (Note $0 \in D_m$, so membership in D_m alone would not exclude $G_c = 0$; this lemma does.) □

6 Homogenization: why the degree must be m

Main message

A size-m affine determinant homogenizes to a degree-m determinant. Homogenizing the target family at any smaller degree would break the identity with the homogenized matrix. At the special fibre the only extra factor is x_0^{m-n}, and the Fermat cone still occurs with multiplicity one.

Example 6.1 (What padding looks like when $m = n + 2$)**.** At the special parameter,

$$\widehat{G}_0 = x_0^2 \widetilde{F}_n.$$

The hyperplane $x_0 = 0$ occurs with multiplicity two and is invisible to the open differential graph at its generic points. The Fermat cone $V(\widetilde{F}_n)$ occurs with multiplicity one and is visible. The factorization therefore creates non-reduced padding without weakening the coefficient of the component used in the lower bound.

Lemma 6.1. $m \ge n$.

Degree is already a closed obstruction

Every polynomial in D_m has degree at most m, and the subspace of polynomials of degree at most m is closed. Therefore a degree-n target cannot lie in $\overline{D_m}$ when $m < n$. The coefficient-function proof below spells this out inside the chosen curve family.

Proof. If $m < n$, then $\deg_x G_c \le m < n$ for every $c \in C$ — the coefficient function of the monomial x_1^n is identically zero on C — contradicting $G_0 = F_n$. □

Definition 6.2. $\widehat{G} := x_0^m\, G(x/x_0) \in \mathcal{O}(C)[x_0, \dots, x_n]_m$, the *degree-m homogenization* of the family. Its coefficients are a reindexing of the coefficients of G, hence regular on C, and in the chart $x_0 = 1$ one has $\widehat{G}_c = G_c$.

Lemma 6.3 (Fiberwise determinantal identity). *For $c \in C^\circ$: $\widehat{G}_c = \det \widehat{A}_c$, where $\widehat{A}_c(x_0, x) := x_0 A_c(x/x_0)$ is an $m \times m$ matrix of linear forms on $\mathbb{P}^n$ (symmetric if A_c is).*

Proof. $\det \widehat{A}_c = \det\big(x_0 A_c(x/x_0)\big) = x_0^m \det A_c(x/x_0) = x_0^m G_c(x/x_0) = \widehat{G}_c$. □

Remark 6.1 (The homogenization degree). The determinant of the homogenized matrix is the *degree-m* homogenization of G_c. If $d = \deg_x G_c < m$, the degree-d homogenization differs by the factor x_0^{m-d}:

$$\det \widehat{A}_c = x_0^{m-d} \widehat{G}_c^{(d)}.$$

The extra hyperplane component has conormal $V(x_0) \times \{[1 : 0 : \cdots : 0]\}$, whose multidegrees vanish in every slot $k \ge 1$. Thus either convention gives the same δ_{n-2} for $n \ge 3$, but degree-m homogenization keeps the determinantal identity exact and is used throughout.

Lemma 6.4 (The special fibre). $\widehat{G}_0 = x_0^{m-n}\, \widetilde{F}_n$, *and the Fermat cone $X = V(\widetilde{F}_n)$ is a* multiplicity-one *component of the divisor $V(\widehat{G}_0)$.*

Roadmap: separate the padding factor from the target factor

First compute the special homogeneous form. Then check three facts: $\widetilde{F}_n$ is irreducible, x_0 does not divide it, and its exponent in the factorization is one. These facts locate every extra multiplicity on the hyperplane at infinity rather than on the Fermat cone.

Proof. $\widehat{G}_0 = x_0^m F_n(x/x_0) = x_0^{m-n} \widetilde{F}_n$ since F_n is homogeneous of degree n and $m \ge n$ (Lemma 6.1). The form $\widetilde{F}_n$ is irreducible: the hypersurface $Z = V(F_n) \subset \mathbb{P}^{n-1}$ is smooth (Section 9) and connected (a hypersurface in $\mathbb{P}^{n-1}$, $n - 1 \ge 2$), hence irreducible, so its squarefree defining form is irreducible, and the cone has the same defining form. Also $x_0 \nmid \widetilde{F}_n$. Hence the irreducible factorization of $\widehat{G}_0$ is $x_0^{m-n} \cdot \widetilde{F}_n$, and the Fermat component carries multiplicity exactly one: all padding multiplicity sits on the hyperplane at infinity, none on the Fermat component, whatever $m - n \ge 0$ is. □

7 Conservation in a family of Gauss graphs

The specialization principle

Take the components of the open Gauss graph that dominate the parameter curve and close them up. The resulting family $W \to C$ is flat. Therefore its fibre cycle is effective and its multidegrees are constant. The special fibre may acquire extra components, but they carry nonnegative coefficients and can only add nonnegative contributions to the special-fibre total.

The construction of W has four operations, each serving a different purpose:

(1) form the open graph where $\widehat{G}_c = 0$ and $d\widehat{G}_c \neq 0$;
(2) decompose that graph into irreducible components;
(3) discard components whose image in the parameter curve is finite ("vertical" components); and
(4) close the remaining components in the projective ambient space over the whole curve, including the special parameter 0.

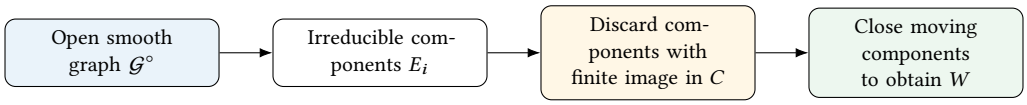


Fig. 2. Construction of the moving Gauss-graph family. Only components that occur for a dense set of parameters are retained.

For $c \in U$ set $\Gamma_c := \Gamma_1(\widehat{G}_c)$, the cycle of Definition 3.1 for the fibre. Define the locally closed algebraic set

$$\mathcal{G}^\circ := \left\{(x, \xi, c) \in \mathbb{P}^n \times (\mathbb{P}^n)^\vee \times C^\circ \;:\; \widehat{G}_c(x) = 0,\ d\widehat{G}_c(x) \neq 0,\ \xi = [d\widehat{G}_c(x)]\right\},$$

with the reduced structure: it is the graph of the morphism $(x, c) \mapsto [d\widehat{G}_c(x)]$ over the open subset $\{d\widehat{G} \neq 0\}$ of the closed set $\{\widehat{G} = 0\} \subset \mathbb{P}^n \times C^\circ$, the coefficients of $\widehat{G}$ and its partials being regular in c.

Lemma 7.1 (The open graph is smooth over the curve). *The projection $\mathcal{G}^\circ \to C^\circ$ is a smooth morphism of relative dimension $n-1$. In particular $\mathcal{G}^\circ$ is reduced, and its fibres are smooth of pure dimension $n-1$.*

Roadmap: why the open graph is smooth

Forget the dual coordinate for a moment. The open zero set $Z^\circ = \{\widehat{G} = 0,\ d_x\widehat{G} \neq 0\}$ is a relative hypersurface whose derivative in the projective variables is nonzero. The Jacobian criterion makes Z° smooth, and the projection to the curve has surjective differential. Adding the tangent covector is merely taking the graph of a morphism, which preserves smoothness.

Proof. $\mathcal{G}^\circ$ is the image of

$$Z^\circ := \left\{(x, c) \in \mathbb{P}^n \times C^\circ \;:\; \widehat{G}_c(x) = 0,\ d_x\widehat{G}_c(x) \neq 0\right\}$$

under the graph embedding $(x, c) \mapsto (x, [d_x\widehat{G}_c(x)], c)$, an isomorphism onto $\mathcal{G}^\circ$ over C° (it is the graph of a morphism $Z^\circ \to (\mathbb{P}^n)^\vee$); so it suffices to prove $Z^\circ \to C^\circ$ smooth. Work near a point $(x^*, c^*) \in Z^\circ$ in an affine chart of $\mathbb{P}^n$ and a local coordinate t at c^*. There Z° is the zero locus of the single regular function $\widehat{G}(x, t)$, whose differential at (x^*, c^*) is nonzero — already its x-part $d_x\widehat{G}_{c^*}(x^*)$ is — so Z° is smooth of dimension n at the point, with tangent space $\ker d\widehat{G}$. The kernel of the differential of the projection $\pi : Z^\circ \to C^\circ$ at the point is

$$T_{(x^*,c^*)}Z^\circ \cap \{dt = 0\} = \left\{\xi \;:\; d_x\widehat{G}_{c^*}(x^*) \cdot \xi = 0\right\},$$

of dimension $n-1$; hence $d\pi$ has rank one and is surjective onto $T_{c^*}C^\circ$. A morphism of smooth complex varieties whose differential is surjective at every closed point is a smooth morphism, here of relative dimension $n-1$. □

Lemma 7.2 (Construction, domination, flatness). *$\mathcal{G}^\circ$ has finitely many irreducible components $E_1, \dots, E_r$. After enlarging Σ by finitely many points, the reduced closed set*

$$W \;:=\; \bigcup_{i:\ \pi(E_i)\ \text{dense in}\ C} \overline{E_i} \;\subseteq\; \mathbb{P}^n \times (\mathbb{P}^n)^\vee \times C$$

(closure in the ambient, π the projection to C) satisfies: every irreducible component of W dominates C; W is pure of dimension n; and $\pi : W \to C$ is flat.

Roadmap: why domination gives flatness over a curve

After vertical components are removed, every irreducible component of W maps densely to C. At a point $c \in C$, let t_c be a local parameter. Dominance says that t_c does not vanish identically on any component. Since W is reduced, this means t_c is a non-zero-divisor. Thus each local ring of W is torsion-free over the discrete valuation ring $\mathcal{O}_{C,c}$, and a torsion-free module over a DVR is flat. The full proof also checks purity of dimension.

Proof. Noetherianity gives finitely many E_i. Each image $\pi(E_i) \subseteq C^\circ$ is constructible (Chevalley) in a curve, hence finite or cofinite; put the points of the finite images into Σ, so that over U the fibres of $\mathcal{G}^\circ$ and of $\bigcup_{\text{dom}} E_i$ agree. (Note that the closure of a non-dominating component lies over the same finite image, hence over Σ.) Every component of W is some $\overline{E_i}$ with dense image, and closure preserves domination. Each dominating E_i has fibres of pure dimension $n-1$ over a dense subset of C° (Lemma 7.1), so $\dim \overline{E_i} = n$ and W is pure of dimension n. Flatness: W is reduced, so the zero-divisors of $\mathcal{O}_W$ are the functions vanishing on a component; for a closed point $c \in C$ with uniformizer $t_c \in \mathcal{O}_{C,c}$ (C smooth, so this local ring is a DVR), $\pi^\# t_c$ vanishes on no component (each dominates), hence is a non-zero-divisor in $\mathcal{O}_{W,w}$ for every w over c; thus $\mathcal{O}_{W,w}$ is a torsion-free, hence flat, $\mathcal{O}_{C,c}$-module, and flatness is local [5, III.9.7]. □

Lemma 7.3 (Generic fibres are the Gauss-graph cycles, with coefficient one). *After a further finite enlargement of Σ: for every $c \in U$ the scheme-theoretic fibre W_c has support* $\operatorname{supp}\Gamma_c$, *and $[W_c] = \Gamma_c$ as cycles, with every coefficient equal to one.*

Roadmap: why the generic fibre has exactly coefficient one

Two facts must be checked separately. First, no entire fibre component is created by the boundary of a moving component; deleting finitely many parameters ensures that the open graph is dense in every generic fibre component. Second, at the generic point of such a component the map to C is smooth, so the fibre is reduced there. The cycle coefficient is the length of that generic local ring and is therefore one.

Proof. *Supports.* Fix a dominating E_i. By flatness over the smooth curve, every component of every fibre $(\overline{E_i})_c$ has dimension exactly $n-1$ [5, III.9.5–9.6]. The boundary $\overline{E_i} \smallsetminus E_i$ is closed of dimension $\le n-1$, so over a cofinite set of c its fibres have dimension $\le n-2$; enlarge Σ accordingly. Then no fibre component of $(\overline{E_i})_c$ lies in the boundary, i.e. $(E_i)_c$ is dense in $(\overline{E_i})_c$ and $(\overline{E_i})_c = \overline{(E_i)_c}$. Since the non-dominating components of $\mathcal{G}^\circ$ and their closures lie over Σ, while $\bigcup_i (E_i)_c = \mathcal{G}_c^\circ$ for $c \in U$, summing over the dominating i gives

$$\operatorname{supp} W_c \;=\; \bigcup_i \overline{(E_i)_c} \;=\; \overline{\mathcal{G}_c^\circ} \;=\; \operatorname{supp}\Gamma_c.$$

Coefficients. Enlarge Σ further so that for $c \in U$ the fibres of the pairwise intersections $\overline{E_i} \cap \overline{E_j}$ ($i \neq j$, both dominating), each of dimension $\le n-1$, have dimension $\le n-2$. Let T be an irreducible

component of W_c; by the support computation, T is a component of $\overline{(E_i)_c}$ for some dominating i, and by the dimension counts above T contains a point p lying in $(E_i)_c$, outside the boundary $\overline{E_i} \smallsetminus E_i$, and outside every $\overline{E_j}$ with $j \neq i$. Near p the scheme W coincides with $\overline{E_i}$ and contains the subset E_i, which is open in $\overline{E_i}$ near p (as $\mathcal{G}^\circ$ is open in its closure and the other components of $\mathcal{G}^\circ$ have been avoided); on this open set the projection to C is smooth (Lemma 7.1). Hence $W \to C$ is smooth at p, so the scheme-theoretic fibre W_c is smooth — in particular reduced — at p. Reducedness passes to generizations: $\mathcal{O}_{W_c,\eta_T}$ is a localization of $\mathcal{O}_{W_c,p}$, and localizations of reduced rings are reduced. So W_c is reduced at the generic point of T, and the coefficient of T in the fibre cycle $[W_c]$ — the length of that local ring — equals one. Since the supports agree and Γ_c carries every component with coefficient one by definition, $[W_c] = \Gamma_c$. □

Lemma 7.4 (Conservation of multidegree). *$[W_0]$ is an effective cycle of pure dimension $n-1$, and for every k and every $c \in U$,*

$$\delta_k\big([W_0]\big) = \delta_k\big(\Gamma_c\big).$$

Roadmap: from flatness to conservation of every multidegree

The special fibre has dimension $n-1$ and positive generic-length multiplicities. Proper flatness keeps the Euler characteristic of every bidegree twist $\mathcal{O}_{W_c}(a,b)$ constant. The highest-degree coefficients of this two-variable numerical polynomial are precisely the multidegrees of the fibre cycle, with its scheme-theoretic multiplicities.

Proof. Put $d = n-1$. Each irreducible component of W has dimension n and dominates C. A fibre is cut out locally by a nonzero parameter on the smooth curve, so every irreducible component of its support has dimension d. Its coefficient in the fundamental fibre cycle is the positive finite length of the fibre's local ring at the generic point. This proves purity and effectivity, without requiring the special fibre to be reduced.

For conservation, let $\mathcal{L}_1$ and $\mathcal{L}_2$ be the restrictions to W of $\mathcal{O}(1,0)$ and $\mathcal{O}(0,1)$ from the fixed biprojective ambient. The morphism $\pi : W \to C$ is proper and of finite presentation, and is flat by Lemma 7.2. Every invertible sheaf $\mathcal{L}_1^{\otimes a} \otimes \mathcal{L}_2^{\otimes b}$ is therefore flat over C. Constancy of Euler characteristic in a proper flat family [17, Tag 0B9T] implies that

$$\chi\big(W_c, \mathcal{O}_{W_c}(a,b)\big)$$

is independent of c, for every pair of integers a, b: the curve C is irreducible, hence connected.

For each c, these values are represented by a numerical polynomial $P_c(a,b) \in \mathbb{Q}[a,b]$ of total degree at most d [17, Tag 0BEM]. Equality on all nonnegative integer pairs implies $P_c = P_0$. The homogeneous part of total degree d is

$$(P_c)_d(a,b) = \sum_{k=0}^{d} \frac{\delta_k([W_c])}{(d-k)!\,k!}\, a^{d-k} b^k.$$

This coefficient formula uses the fundamental cycle of the actual scheme-theoretic fibre: its generic local lengths occur as multiplicities [17, Tags 0BEN, 0BEP, 0BES]. Comparing the displayed coefficients gives $\delta_k([W_c]) = \delta_k([W_0])$ for every k. Lemma 7.3 gives $[W_c] = \Gamma_c$ on U, as required. Both twisting variables are needed; a single Hilbert polynomial for one fixed Segre embedding would not by itself determine every multidegree. □

Proof check: the specialization package

At this point the proof has a proper flat family $W \to C$, the generic fibre cycle is exactly Γ_c with coefficient one, and the special fibre is an effective cycle with the same multidegrees. What has *not* yet been proved is that $\mathrm{Con}(X)$ occurs inside $[W_0]$; that geometric containment is the sole task of the next section.

8 Why the Fermat conormal appears in the limit

The only delicate point

It is not enough that tangent hyperplanes have a formal limit: the approximating points must lie on multiplicity-one components, because Γ_1 ignores repeated components. At every smooth affine point of the Fermat cone, a transverse one-variable slice and Hurwitz's theorem produce nearby zeros with nonzero differential. Those zeros are automatically on multiplicity-one branches, and their Gauss points converge to the desired conormal point.

Proof idea. There are three logically distinct tasks.

(1) **Move the zero.** The fixed point x^* need not remain on the nearby hypersurface, so find nearby points x_c with $G_c(x_c) = 0$.

(2) **Keep the differential nonzero.** Show $d\widehat{G}_c(1, x_c) \neq 0$ and identify its projective limit. This places the approximating points in the open Gauss graph, rather than only in its closure.

(3) **Land in the moving family.** Because $c \in U$, the component containing each approximating point cannot be vertical. Its closure is one of the components used to define W, so the limit point lies in W_0.

Fix a smooth affine zero x^* of F_n. Restrict G_c to a line through x^* on which F_n has a simple zero. That zero persists for small c, gradients converge, and Euler's identity controls the homogenizing coordinate of the covector. Closing the dominating Gauss-graph components then places a dense open subset of $\mathrm{Con}(X)$ inside the special fibre W_0.

Proposition 8.1 (Containment). $\mathrm{Con}(X) \subseteq \mathrm{supp}[W_0]$, *and the coefficient of* $\mathrm{Con}(X)$ *in the effective cycle* $[W_0]$ *is at least one.*

The proof must produce points of the *multiplicity-one* Gauss graphs $\Gamma_1(\widehat{G}_c)$ converging into the Fermat conormal; this is the reason for using Γ_1, since the limit divisor $V(\widehat{G}_0) = V(x_0^{m-n}\widetilde{F}_n)$ is badly non-reduced at infinity and nothing guarantees the nearby divisors $V(\widehat{G}_c)$ are reduced either.

Lemma 8.2 (The limit covector at a smooth affine cone point). *Let* $x^* \in V(F_n) \subset \mathbb{C}^n$, $x^* \neq 0$, *so that* $\nabla F_n(x^*) = n\,({x_i^*}^{n-1})_i \neq 0$. *Then*

$$d\widehat{G}_0(1, x^*) \;=\; \big(0,\ \nabla F_n(x^*)\big) \;\neq\; 0,$$

whose projective class $[0 : \nabla F_n(x^*)]$ *is the conormal covector of the cone* X *at the smooth point* $(1, x^*)$.

Roadmap: compute the special tangent covector directly

Differentiate the product $x_0^{m-n}\widetilde{F}_n$. At a point of the Fermat component in the chart $x_0 = 1$, every derivative of the padding factor is multiplied by $\widetilde{F}_n$ and therefore vanishes. The remaining entries are exactly the affine gradient of F_n, with zero in the x_0 coordinate.

Proof. With $\widehat{G}_0 = x_0^{m-n}\widetilde{F}_n$ (Lemma 6.4):

$$\partial_0\widehat{G}_0 = (m-n)\,x_0^{m-n-1}\widetilde{F}_n + x_0^{m-n}\,\partial_0\widetilde{F}_n, \qquad \partial_i\widehat{G}_0 = x_0^{m-n}\,\partial_i\widetilde{F}_n \ \ (i \geq 1).$$

At $(1, x^*)$: the first term of ∂_0 vanishes because $\widetilde{F}_n(1, x^*) = F_n(x^*) = 0$, the second because $\widetilde{F}_n$ does not involve x_0, and $\partial_i\widehat{G}_0(1, x^*) = \partial_i F_n(x^*)$. The hyperplane $\{\sum_{i\geq 1}\partial_i F_n(x^*)\,y_i = 0\}$ is the projective tangent hyperplane of X at $(1, x^*)$ and contains the vertex q_0 — the $\xi_0 = 0$ signature of a cone. Note the mechanism: the non-reduced factor x_0^{m-n} contributes to the differential only through the term carrying $\widetilde{F}_n$ itself, which dies *because the differential is evaluated on the zero locus of* $\widetilde{F}_n$. Padding is differentially invisible along the Fermat component. □

Lemma 8.3 (Hurwitz persistence onto multiplicity-one branches). *For every x^* as in Lemma 8.2 there are parameters $c_k \in U$ with $c_k \to 0$ analytically and points $x_k \to (1, x^*)$ in the chart $x_0 = 1$ such that*

$$\widehat{G}_{c_k}(x_k) = 0, \qquad d\widehat{G}_{c_k}(x_k) \neq 0, \qquad \big[d\widehat{G}_{c_k}(x_k)\big] \longrightarrow \big[0 : \nabla F_n(x^*)\big].$$

Moreover each x_k lies on a multiplicity-one component of $V(\widehat{G}_{c_k})$, so $\big(x_k, [d\widehat{G}_{c_k}(x_k)]\big)$ lies in the open graph $\mathcal{G}^\circ_{c_k}$ *defining Γ_{c_k} — not merely in its closure.*

Roadmap: a one-variable implicit-function argument

Choose a direction w on which the nonzero differential $dF_n(x^*)$ evaluates to one. The restricted function $\varphi_0(s) = F_n(x^* + sw)$ then has a simple zero at $s = 0$. Uniform coefficient convergence gives uniform convergence of φ_c and its derivative on a small disk. Hurwitz's theorem preserves the simple zero, producing $s_c \to 0$. Gradient convergence keeps the full differential nonzero, and Euler's identity supplies the missing homogeneous coordinate.

Proof. Work in the chart $x_0 = 1$, where $\widehat{G}_c = G_c$. Choose $w \in \mathbb{C}^n$ with $\langle dF_n(x^*), w\rangle = 1$ — possible since the functional is nonzero; one may *not* simply take $w = \nabla F_n(x^*)$, which can be isotropic over $\mathbb{C}$. Define $\varphi_c(s) := G_c(x^*+sw)$, holomorphic in s on a disk $\overline{D_\rho}$, with coefficients regular (hence analytically continuous) in c. Then $\varphi_0(s) = F_n(x^* + sw)$ has a simple zero at $s = 0$: $\varphi_0(0) = 0$, $\varphi_0'(0) = 1$. Shrink ρ so that $s = 0$ is the only zero of φ_0 in $\overline{D_\rho}$. Since $G_c \to F_n$ coefficientwise as $c \to 0$ (regular functions on C are analytically continuous and take the value at 0 prescribed by Lemma 5.1(a)), $\varphi_c \to \varphi_0$ uniformly on $\overline{D_\rho}$; by Hurwitz's theorem — if holomorphic $f_k \to f \not\equiv 0$ uniformly on $\overline{D_\rho}$, f zero-free on ∂D_ρ with exactly one zero counted with multiplicity inside, then so is f_k for large k, with its zero converging to that of f (argument principle) — there is, for each c small in U, a unique $s_c \in D_\rho$ with $G_c(x^* + s_c w) = 0$ and $s_c \to 0$. Set $x_c := x^* + s_c w$.

Gradients: $\nabla G_c \to \nabla F_n$ uniformly on compacts (again coefficientwise convergence of polynomials of bounded degree), so $\nabla G_c(x_c) \to \nabla F_n(x^*) \neq 0$, and for small c the affine entries $\partial_i\widehat{G}_c(1, x_c) = \partial_i G_c(x_c)$, $i \geq 1$, are not all zero, whence $d\widehat{G}_c(1, x_c) \neq 0$. The 0-th entry is determined by Euler's relation for the degree-m form $\widehat{G}_c$:

$$\partial_0\widehat{G}_c(1, x_c) = m\,G_c(x_c) - \sum_{i\geq 1} x_{c,i}\,\partial_i G_c(x_c) \ \longrightarrow\ 0 - \langle x^*, \nabla F_n(x^*)\rangle = -n\,F_n(x^*) = 0,$$

so $[d\widehat{G}_c(1, x_c)] \to [0 : \nabla F_n(x^*)]$.

Multiplicity-one membership — the point of the construction: x_c is a point of $V(\widehat{G}_c)$ at which the differential does not vanish. More explicitly, write a local factorization $\widehat{G}_c = u\prod_j g_j^{a_j}$ with u a unit. If the point lies on a factor with $a_j \geq 2$, every term in the derivative contains g_j and vanishes. If it lies on two distinct components, each product-rule term contains the equation of

the other component and again vanishes. Therefore a nonzero differential forces the point to lie on exactly one local component, and that component has multiplicity one. A point lying on a multiplicity-≥ 2 component has $d\widehat{G}_c = 0$ there (Lemma 3.2), so x_c lies only on multiplicity-one components and, having nonvanishing differential, lies in the open graph. No squarefreeness of $\widehat{G}_c$ is assumed; whatever multiple components exist live elsewhere. Finally, restrict the parameters to a sequence $c_k \in U$, $c_k \to 0$, which exists since Σ is finite. □

Roadmap: close a dense set of conormal points

For each smooth affine point of the Fermat cone, Lemma 8.3 produces a convergent sequence in the moving open graph. Closedness of W puts the limit in W_0. These limits cover the conormal graph over a dense open subset of X, so their closure is all of $\mathrm{Con}(X)$. Finally, an irreducible subvariety of the same dimension as the effective fibre cycle is an actual fibre component and therefore has a positive integer coefficient.

OF PROPOSITION 8.1. The points constructed in Lemma 8.3 lie in $\mathcal{G}^\circ_{c_k}$ with $c_k \in U$; the component of $\mathcal{G}^\circ$ carrying such a point dominates C (a non-dominating component has image inside Σ, which U avoids), so the point lies in W. Since W is Zariski-closed, it is closed in the analytic topology, and the analytic limit $\big((1, x^*), [0 : \nabla F_n(x^*)]\big)$ lies in $W \cap \pi^{-1}(0) = W_0$. As x^* ranges over $V(F_n) \setminus \{0\}$, these limit points range over the conormal graph of X over the smooth affine cone points, which is dense in $\mathrm{Con}(X)$: the smooth locus of X is $X \setminus \{q_0\}$ (Proposition 9.1(a)), its intersection with the chart $x_0 \neq 0$ is dense in X (X is irreducible and $X \not\subseteq \{x_0 = 0\}$), and the conormal covector over a smooth point is unique. Hence $\mathrm{Con}(X) \subseteq \mathrm{supp}[W_0]$.

Coefficient: $[W_0]$ is effective of pure dimension $n-1$ (Lemma 7.4); $\mathrm{Con}(X)$ is irreducible of dimension $n-1$ and is contained in the support, hence is an irreducible component of $\mathrm{supp}[W_0]$, and its coefficient is a positive integer. □

Proof check: why this proves component containment, not merely pointwise convergence

The approximating points lie in components that dominate C; hence they are contained in W, not in a discarded vertical component. Their limits range over a dense open part of the Fermat conormal. Since W_0 is closed, it contains the closure of that dense set. Because both $\mathrm{Con}(X)$ and the fibre components have dimension $n-1$, containment in the support makes $\mathrm{Con}(X)$ itself an irreducible component with positive coefficient.

Remark 8.1 (Additional components in the limit). Besides $\mathrm{Con}(X)$, the cycle $[W_0]$ may contain: conormal-type cycles over the hyperplane at infinity (limits of graph points escaping to $x_0 = 0$; for $c \in U$ the divisor $V(\widehat{G}_c)$ may itself contain $V(x_0)$ as a component of multiplicity $m - \deg_x G_c$, which when that multiplicity is one rides through the family with $\delta_{n-2} = 0$, cf. Remark 6.1, and when ≥ 2 is invisible to Γ_1); cycles of the form $S' \times \{\text{linear space}\}$ over singular strata; and Lagrangian-type components over the singular locus of the limit divisor. Effectivity is the key point: extra components enter $[W_0]$ with nonnegative coefficients and nonnegative multidegrees (Lemma 3.5) and can only *add*. Nothing subtracts the $\mathrm{Con}(X)$ summand, because cycle specialization has no negative coefficients and Proposition 8.1 pins $\mathrm{Con}(X)$ inside the support with coefficient ≥ 1. Contrast the two limit objects: the *divisor* $V(\widehat{G}_0)$ is badly non-reduced at infinity; the *cycle* $[W_0]$ is insensitive to that, precisely because Γ_1 is built from the differential, which Lemma 8.2 shows survives along the Fermat component.

Corollary 8.4 (Special ≤ generic). *For every $c \in U$:*

$$\delta_{n-2}\big(\mathrm{Con}(X)\big) \;\le\; \delta_{n-2}\big([W_0]\big) \;=\; \delta_{n-2}(\Gamma_c).$$

Proof. Write $[W_0] = a \cdot \mathrm{Con}(X) + \sum_j a_j V_j$ with $a \ge 1$, $a_j \ge 0$ integers, and V_j irreducible of dimension $n-1$ (Proposition 8.1, Lemma 7.4). Each $\delta_{n-2}(V_j) \ge 0$ (Lemma 3.5), so $\delta_{n-2}([W_0]) \ge a\,\delta_{n-2}(\mathrm{Con}(X)) \ge \delta_{n-2}(\mathrm{Con}(X))$. The equality is Lemma 7.4. □

9 Cones and the next-to-top multidegree

Why the index is $n-2$

Every tangent hyperplane to a cone contains the vertex, so the top dual slot δ_{n-1} vanishes. The next slot adds one generic condition on the point of the cone; this cuts each ruling line once and leaves exactly the dual-degree count of the base hypersurface.

To read $\delta_{n-2}(\mathrm{Con}(X_Z))$ enumeratively, choose

- a generic hyperplane $M \subset \mathbb{P}^n$ in the primal factor; and
- a generic codimension-$(n-2)$ linear space $L \subset (\mathbb{P}^n)^\vee$ in the dual factor.

First impose the dual condition. It selects finitely many tangent hyperplanes to the base Z, in number $\deg Z^\vee$. Over each selected base tangency, the cone still has one ruling line of possible primal points. The hyperplane M intersects that line once. Thus the count is unchanged when passing from the base conormal to the next-to-top conormal multidegree of the cone.

Proposition 9.1 (Cone shift). *Let $Z \subset \mathbb{P}^{n-1} = \{x_0 = 0\}$ be a smooth irreducible hypersurface of degree $e \ge 2$, $n \ge 3$, and let $X_Z \subset \mathbb{P}^n$ be the cone over Z with vertex $q_0 = [1:0:\cdots:0]$. Then:*

(a) $\mathrm{Sing}\, X_Z = \{q_0\}$, *and* $\mathrm{Con}(X_Z) \subset \mathbb{P}^n \times q_0^\perp$, *where* $q_0^\perp = \{\xi_0 = 0\} \cong (\mathbb{P}^{n-1})^\vee$;
(b) $\delta_{n-1}(\mathrm{Con}(X_Z)) = 0$ *(the cone is dual-deficient);*
(c) $\delta_{n-2}(\mathrm{Con}(X_Z)) = \deg Z^\vee$.

In particular, for the Fermat cone X ($Z = V(F_n)$ smooth since ∇F_n vanishes only at the origin, $e = n$), Lemma 3.7 with $N = n-1$ gives

$$\delta_{n-2}\big(\mathrm{Con}(X)\big) \;=\; n(n-1)^{n-2}.$$

Roadmap: prove the cone shift in three geometric moves

Part (a) identifies the smooth locus and shows that every tangent hyperplane contains the vertex. Part (b) uses that restriction to explain why the top dual multidegree vanishes. Part (c) imposes the next-to-top flag: the dual conditions count tangent hyperplanes to the base, and the remaining primal condition cuts each ruling line once.

Proof. Write the defining form of X_Z as $q(x_1, \ldots, x_n)$, independent of x_0. (a) Its differential is $(0, \nabla q)$. Smoothness of Z shows that this differential is nonzero at every point of X_Z except q_0; at q_0 it vanishes because $e \ge 2$. The conormal covector at $[x_0 : z]$ with $z \ne 0$ is $[0 : \nabla q(z)]$. Thus every tangent hyperplane contains q_0, and taking closures gives $\mathrm{Con}(X_Z) \subset \mathbb{P}^n \times q_0^\perp$.

Radial projection

$$\rho : X_Z \smallsetminus \{q_0\} \longrightarrow Z, \qquad [x_0 : z] \longmapsto [z],$$

is a morphism. On this open set the conormal is the graph of the displayed covector, and ρ identifies its dual coordinate with the Gauss covector of Z. This description is used only away from the vertex; no global bundle map from $\mathrm{Con}(X_Z)$ to $\mathrm{Con}(Z)$ is required. The open graph has dimension $(n-2)+1 = n-1$, and so does its closure.

(b) The dual image is $Z^\vee \subset q_0^\perp$, which has dimension $n-2$ by Lemma 3.7. A generic collection of $n-1$ dual hyperplanes misses this image. Hence $\delta_{n-1}(\mathrm{Con}(X_Z)) = 0$.

(c) Choose a generic primal hyperplane M avoiding q_0. Write its equation as $\alpha x_0 + \sum_i b_i x_i = 0$, with $\alpha \neq 0$. Radial projection restricts to an isomorphism

$$\rho : X_Z \cap M \xrightarrow{\sim} Z, \qquad \rho^{-1}([z]) = \left[-\alpha^{-1}\sum_i b_i z_i : z\right].$$

All of this section lies in the smooth locus of the cone. On the conormal graph, the equation of M cuts the ruling direction with nonzero coefficient α; therefore it cuts with multiplicity one. Consequently the scheme-theoretic section

$$\mathrm{Con}(X_Z) \cap \left(M \times (\mathbb{P}^n)^\vee\right)$$

is isomorphic to $\mathrm{Con}(Z)$, preserving the dual coordinate under $q_0^\perp \cong (\mathbb{P}^{n-1})^\vee$.

Now impose $n-2$ generic dual hyperplanes. Their restrictions to $q_0^\perp$ are generic hyperplanes there. By the preceding isomorphism, the resulting intersection number is

$$\delta_{n-2}(\mathrm{Con}(X_Z)) = \deg\left(\mathrm{Con}(Z)\cdot h_2^{n-2}\right) = \deg Z^\vee.$$

For the last equality, the conormal projection to $Z^\vee$ is birational in characteristic zero by Lemma 3.7; a generic line in $(\mathbb{P}^{n-1})^\vee$ meets its hypersurface image in $\deg Z^\vee$ points, each with one conormal lift. Generic transversality as in Lemma 3.5 also realizes these intersection numbers as reduced point counts. This proves (c). □

Remark 9.1. Slot $n-1$ is blind (part (b)); slot $n-2$ is the first informative multidegree of a cone and records exactly the dual degree of the base. This is the global avatar of the companions' local invariant — the top polar degree of the projectivized tangent cone at the vertex *is* $\deg Z^\vee$ — and it is the conversion of that local invariant into a global multidegree, available only because F_n is homogeneous, that lets conservation (Lemma 7.4) act on it. For $n = 3$, the identity gives $3 \cdot 2 = 6$, the classical dual degree of a smooth plane cubic.

10 Combining the lower and upper counts

The entire numerical comparison

For a generic determinantal fibre Γ_c,

$$n(n-1)^{n-2} = \delta_{n-2}(\mathrm{Con}(X)) \le \delta_{n-2}(\Gamma_c) \le B(m,n) \le \left(\frac{4em}{n-1}\right)^{n-1}.$$

The symmetric proof replaces the last term by $\left(\frac{2em}{n-1}\right)^{n-1}$. Taking $(n-1)$-st roots gives the stated quadratic lower bounds.

Lemma 10.1 (Estimates). *For $n \ge 3$ and $m \ge n$:*

$$B(m,n) \;\le\; 2^{n-2}\binom{2m-1}{n-1} \;\le\; \left(\frac{4em}{n-1}\right)^{n-1}, \qquad 2^{n-2}\binom{m}{n-1} \;\le\; \left(\frac{2em}{n-1}\right)^{n-1}.$$

Roadmap: reduce the exact coefficient to a standard binomial estimate

The exact sum $B(m,n)$ contains a factor $\binom{n-2}{i-1}$. Bound this factor by the sum of all binomial coefficients, 2^{n-2}. The remaining convolution is a Vandermonde coefficient $\binom{2m-1}{n-1}$. Finally apply $\binom{N}{k} \le (eN/k)^k$ and make only the two harmless relaxations $2m-1 \le 2m$ and $2^{n-2} \le 2^{n-1}$.

Proof. $\binom{n-2}{i-1} \le 2^{n-2}$, and by Vandermonde $\sum_i \binom{m}{i}\binom{m-1}{n-1-i} = \binom{2m-1}{n-1}$, of which $B(m,n)/2^{n-2}$ retains a sub-sum of nonnegative terms. Then $\binom{N}{k} \le (eN/k)^k$ gives $2^{n-2}\binom{2m-1}{n-1} \le 2^{n-2}\big(\frac{e(2m-1)}{n-1}\big)^{n-1} \le 2^{n-1}\big(\frac{2em}{n-1}\big)^{n-1} = \big(\frac{4em}{n-1}\big)^{n-1}$, and similarly $2^{n-2}\binom{m}{n-1} \le 2^{n-1}\big(\frac{em}{n-1}\big)^{n-1} = \big(\frac{2em}{n-1}\big)^{n-1}$. □

Trace of every comparison in the main chain

$n(n-1)^{n-2} = \delta_{n-2}(\mathrm{Con}(X))$	Cone-shift identity and Fermat dual degree.
$\delta_{n-2}(\mathrm{Con}(X)) \le \delta_{n-2}(\Gamma_c)$	$\mathrm{Con}(X)$ is an effective component of $[W_0]$, and multidegrees are conserved.
$\delta_{n-2}(\Gamma_c) \le B(m,n)$	Determinantal conormal bound applied to the exact generic fibre.
$B(m,n) \le (4em/(n-1))^{n-1}$	Lemma 10.1.
$(n-1)^{n-1} \le n(n-1)^{n-2}$	The elementary inequality $n \ge n-1$.

of Theorem 1.1. Suppose $\overline{\mathrm{dc}}(F_n) \le m$. Run Lemmas 5.1–6.4: a smooth curve C, the family $\widehat{G}$ homogenized at degree m (so $m \ge n$ by Lemma 6.1), fiberwise identities $\widehat{G}_c = \det \widehat{A}_c$ on C°, special fibre $x_0^{m-n}\widetilde{F}_n$ with multiplicity-one Fermat component. Build the family W and fix any $c \in U$ (U is infinite, so such c exists). Chain the results:

$$n(n-1)^{n-2} \overset{9.1}{=} \delta_{n-2}\big(\mathrm{Con}(X)\big) \overset{8.4}{\le} \delta_{n-2}\big(\Gamma_c\big) \overset{1.4(\mathrm{i})}{\le} B(m,n) \overset{10.1}{\le} \left(\frac{4em}{n-1}\right)^{n-1},$$

the middle application of Theorem 1.4(i) being directly: $\Gamma_c = \Gamma_1(\det \widehat{A}_c)$ with $\det \widehat{A}_c = \widehat{G}_c \ne 0$ (Lemmas 6.3 and 5.2), and the theorem's only hypothesis is exactly that. Now $n(n-1)^{n-2} \ge (n-1)\cdot(n-1)^{n-2} = (n-1)^{n-1}$, so taking $(n-1)$-st roots of $(n-1)^{n-1} \le \big(\frac{4em}{n-1}\big)^{n-1}$ gives $n-1 \le \frac{4em}{n-1}$, i.e. $m \ge \frac{(n-1)^2}{4e}$. Every use of a generic choice occurred after the fixed fibre $\widehat{A}_c$ was selected; no genericity of the determinantal representation itself is required. □

of Theorem 1.2. Identical, with D_m^{sym} in Lemma 5.1 (the image of the symmetric matrix space is likewise irreducible constructible, and curve selection is representation-agnostic), symmetric fiberwise representations A_c, $\widehat{A}_c$ symmetric (homogenization preserves symmetry, Lemma 6.3), Theorem 1.4(ii) in the chain, and the second estimate of Lemma 10.1: $(n-1)^{n-1} \le \big(\frac{2em}{n-1}\big)^{n-1}$, so $m \ge \frac{(n-1)^2}{2e}$. Sections 7 –9 are representation-agnostic — they see only the polynomial family — and need no change. The model is non-vacuous: $\overline{\mathrm{sdc}}(F_n) \le \mathrm{sdc}(F_n) \le 2n^2 + 2n + 1$ [14, Prop. 5.1]. □

Remark 10.1 (No hidden logarithmic loss; where the slack is). The bounds $(n-1)^2/(4e)$ and $(n-1)^2/(2e)$ hold as stated for every $n \ge 3$. In the form $(\frac{1}{4e} - o(1))n^2$, the $o(1)$ is only the identity $(n-1)^2/n^2 = 1 - O(1/n)$. The bounds are not the exact thresholds. Let $m^*(n)$ be the least m with $n(n-1)^{n-2} \le B(m,n)$. Then $m^*(n) = \big(1 + O(\frac{\log n}{n})\big)\,(n-1)^2/(4e)$; at $n = 320$, for example, $m^*(n) = 9676$ while $(n-1)^2/(4e) \approx 9359$, a ratio of 1.034. The symmetric threshold behaves the same way (ratio 1.023 at $n = 320$). Nearly all of this slack comes from Lemma 10.1: the estimate $\binom{N}{k} \le (eN/k)^k$ drops a Stirling factor of order $\sqrt{k}$, and the relaxations of $B(m,n)$ to $2^{n-2}\binom{2m-1}{n-1}$ and of 2^{n-2} to 2^{n-1} drop further polynomial factors. After the $(n-1)$st root, each of these costs a factor $1 + O(\frac{\log n}{n})$. By contrast, the absorptions $n(n-1)^{n-2} \ge (n-1)^{n-1}$ and $2m - 1 \le 2m$ cost only a factor $(n/(n-1))^{1/(n-1)} = 1 + O(1/n^2)$ and an additive $\frac{1}{2}$. The exact thresholds therefore approach $1/(4e) = 0.09197\ldots$ and $1/(2e) = 0.18394\ldots$, in units of $(n-1)^2$, from above.

of Corollary 1.3. $\mathrm{dc} \geq \overline{\mathrm{dc}}$ and $\mathrm{sdc} \geq \overline{\mathrm{sdc}}$ by definition. Alternatively and more directly, Remark 4.3 derives the exact inequalities from Theorem 1.4 and Proposition 9.1 alone, with no specialization machinery. □

11 What the argument uses, and what it does not

The proof is robust to singular and non-reduced determinantal fibres, but it is not a universal lower-bound machine. The following list separates genuine hypotheses from technical conveniences.

(1) Γ_1 *throughout.* Every fiberwise object is the multiplicity-one Gauss-graph cycle, never the full reduced conormal of the fibre. Theorem 1.4 bounds exactly Γ_1; whether the reduced conormal of a non-squarefree determinant obeys the same bound is open (Remark 4.4) and is not needed.

(2) *Why the Fermat conormal survives.* Smooth affine zeros of F_n persist as zeros of G_c with nonvanishing differential (Hurwitz plus uniform C^1 convergence), and a nonvanishing differential forces membership in multiplicity-one components; so the approximating points are honest open-graph points, and properness pushes their limits into W_0. The padding x_0^{m-n} is differentially dead along the Fermat component (Lemma 8.2) and contributes only additional effective components (Remark 8.1).

(3) *No genericity, and no compatibility, of the representations.* The A_c are arbitrary, and no relation among them across different c is assumed; the only object with structure along the parameter is the polynomial family G. The only generic choices in the paper are flags (Lemma 3.5, used in Theorem 1.4 and Proposition 9.1) and the constant matrix Λ of Step 3, all chosen *after* the data.

(4) *Repeated factors are allowed everywhere.* Multiplicity-≥ 2 components of any fibre, including a hyperplane at infinity of multiplicity $m - \deg_x G_c$, are invisible to Γ_1 on both sides of the chain (Remarks 4.2, 6.1, 8.1); a multiplicity-one hyperplane at infinity has $\delta_{n-2} = 0$ and is harmless.

(5) *Transversality and flatness inventory.* Kleiman transversality: Sections 4 and 9. Fulton positivity / localized Chern classes: Lemma 3.6. Curve selection: Lemma 5.1. Smoothness of the open graph over the curve (Jacobian criterion) and reducedness of the generic fibre cycles: Lemmas 7.1 and 7.3. Torsion-freeness over a DVR and flat-fibre dimension theory: Lemmas 7.2 and 7.3. Specialization of cycle classes: Lemma 7.4. Hurwitz: Lemma 8.3.

(6) *Homogeneity is essential.* The cone-shift identity consumes homogeneity of the target: it converts the vertex-local invariant into the global multidegree that specialization conserves. For an inhomogeneous target no such bridge is provided, and the method as written says nothing.

(7) *No permanent claim.* For the permanent the special-side input would be a lower bound on the relevant conormal multidegree of the (singular) permanent hypersurface, which we do not know; the transfer machinery of Sections 5 –8 would apply once such a bound exists, since nothing there is specific to F_n except homogeneity and Lemma 8.2's use of the explicit gradient (which generalizes to any reduced homogeneous target along its smooth affine locus).

(8) *Base changes and shrinkings.* One normalization (Lemma 5.1) and finitely many enlargements of the excluded set Σ (Lemmas 5.1, 5.2, 7.2, 7.3). None alters the hypothesis $F_n \in \overline{D_m}$ or the parameter m; the final chain needs a single $c \in U$, and U is infinite.

(9) *Characteristic zero.* Used in Kleiman transversality, reflexivity (Lemma 3.7), the symmetric quadratic-form argument in Step 4 of Section 4.4, and the analytic Hurwitz argument (replaceable by a Newton–Puiseux argument over $\mathbb{C}[[t]]$, which we do not pursue). The statement itself fails as a uniform theorem in positive characteristic: for $\mathrm{char} = p > 2$ and $n = p^r$ the Frobenius identity makes F_n a perfect power of a linear form, with $\mathrm{sdc}(F_n) \leq n$ [14, Rem. 6.1].

Open questions. (1) Improve the constants toward the upper bounds; the Bézout count is an overcount and the limit cycle $[W_0]$ carries identifiable additional components (Remark 8.1) whose multidegrees could in principle be subtracted. (2) Bound the relevant conormal multidegree of the permanent hypersurface from below, activating item 7. (3) Settle Remark 4.4. (4) Is $\overline{\mathrm{dc}}(F_n) < \mathrm{dc}(F_n)$ for some n? Both are now pinned to $\Theta(n^2)$ with the same lower-bound constant, and the question of a genuine border gap for this family becomes meaningful.

A Glossary for readers outside algebraic geometry

This appendix records the precise informal meaning of the terms that recur in the proof. The formal definitions and citations remain in the main text.

Affine chart The subset $x_0 \neq 0$ of projective space, identified with ordinary affine space by setting $x_0 = 1$.

Conormal variety The closure of pairs (x, H) where x is a smooth point of a variety and H is a hyperplane containing the tangent space at x. For a hypersurface, H is represented by the projective gradient.

Cycle A formal integer combination of irreducible subvarieties of the same dimension. The coefficient records geometric multiplicity.

Effective cycle A cycle all of whose coefficients are nonnegative. This property makes multidegree inequalities monotone componentwise.

Dominating component An irreducible component whose image is dense in the parameter curve. Such a component occurs for a moving set of fibres.

Vertical component A component whose image is not dense; over a curve, its image is finite after constructibility is used.

Flat family A family with no algebraic torsion in the parameter. In this paper, reducedness plus domination over a smooth curve makes the local rings torsion-free over a DVR and hence flat. Flatness is used to specialize fibre cycles with multiplicities preserved.

Generic choice A choice outside a proper Zariski-closed exceptional set. Only auxiliary flags and the map Λ are chosen generically; the input determinant is arbitrary.

Gauss graph The graph of the map sending a smooth hypersurface point to its tangent hyperplane.

Multiplicity-one component An irreducible factor occurring to exponent one in the defining equation. At its generic smooth points the differential is nonzero.

Multidegree An intersection number in a product of projective spaces. Here δ_k counts $n-1-k$ generic primal conditions and k generic dual conditions.

Properness The algebro-geometric compactness property. A closed subvariety of a projective bundle over the parameter curve is proper over the curve, so limits remain in the total space.

Reduced isolated point
A zero-dimensional local component whose local ring is $\mathbb{C}$; its intersection multiplicity is one.

Scheme-theoretic fibre
The fibre defined by the parameter equation in the structure sheaf, including nilpotents and multiplicities rather than only the underlying point set.

Specialization of cycles
The passage from a generic fibre cycle to the cycle of a special fibre, with its local-length multiplicities. In our proper flat family in a fixed product of projective spaces, every multidegree is conserved; Lemma 7.4 proves this using all bidegree twists.

B Standard geometric inputs and exactly where they are used

Input	Statement used	Location and role
Curve selection	A closure point of a constructible set lies on an algebraic curve meeting the set densely	Lemma 5.1; replaces abstract border membership by a one-parameter polynomial family
Normalization of a curve	The normalized curve is smooth, with DVR local rings at closed points	Lemma 5.1 and Lemma 7.2; supplies a regular parameter and the torsion-free criterion
Kleiman transversality	A generic translate meets prescribed varieties transversely in their chosen dense opens	Step 1 of Theorem 1.4 and Proposition 9.1; converts multidegrees into reduced point counts
Localized top Chern class	Isolated zeros of a section consume positive local multiplicity bounded by the top Chern number	Lemma 3.6; permits arbitrary excess components in the kernel incidence
Torsion-free over a DVR	A torsion-free module over a DVR is flat	Lemma 7.2; turns domination of reduced components into flatness
Euler characteristic and numerical intersections	Proper flatness keeps every twist's Euler characteristic constant; the top coefficients give multiplicity-weighted intersection numbers	Lemma 7.4; apply both twisting variables to conserve every multidegree
Hurwitz / argument principle	A simple zero of a holomorphic function persists under uniform perturbation	Lemma 8.3; moves smooth Fermat zeros to nearby fibres
Projective duality	A smooth degree-e hypersurface has dual degree $e(e-1)^{N-1}$	Lemma 3.7 and Proposition 9.1; computes the target-side count

C Proof audit: no hidden transfer step

The following checklist gives a direct route for verifying the logical closure of the argument.

(1) Starting from $\overline{\mathrm{dc}}(F_n) \le m$, Lemma 5.1 produces a regular polynomial family. It does not require a regular family of matrices.

(2) Lemma 6.3 applies Theorem 1.4 to each chosen generic fibre separately; hence the absence of a coherent matrix family causes no gap.

(3) Degree-m homogenization gives the exact determinant identity and the special factorization $x_0^{m-n}\widetilde{F}_n$.

(4) The open Gauss graph uses only points with nonzero differential, so its fibre closure is exactly the multiplicity-one cycle Γ_c.

(5) Vertical components are removed before closure. Every retained component dominates the curve, making the reduced total space flat.

(6) Generic fibre coefficients are one because the total space is smooth at a dense point of each generic fibre component.

(7) The special fibre is effective, and proper flat specialization conserves its multidegrees.

(8) Lemma 8.3 supplies actual open-graph points on nearby fibres; it does not merely produce formal limiting covectors.

(9) Nonvanishing differential forces those nearby points onto a unique multiplicity-one branch.

(10) Their components dominate the curve because the parameters lie in U; therefore their limits lie in W_0.

(11) Density and equal dimension make $\mathrm{Con}(X)$ a component of $[W_0]$ with positive coefficient.

(12) Effectivity yields $\delta_{n-2}(\mathrm{Con}(X)) \le \delta_{n-2}([W_0])$, and conservation identifies the right side with $\delta_{n-2}(\Gamma_c)$.

(13) Theorem 1.4 bounds that generic-fibre multidegree without smoothness or reducedness assumptions.

(14) Proposition 9.1 and Lemma 3.7 compute the special-side value, and Lemma 10.1 completes the arithmetic.

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D End-to-end Lean verification and reproduction guide

This appendix describes the accompanying Lean development, identifies the precise statements that it proves, and gives instructions for rebuilding and inspecting the proofs. The author of the development is Karthik Sheshadri. The sources and the archived release, which also contains this paper, are at

- Lean sources: `https://github.com/karthik202321/power-sum-dc-lean`;
- archived release: `https://doi.org/10.5281/zenodo.23274251`.

The entry point is `OriginalConormal.lean`; the final numerical theorems are in `Sheshadri/ConormalExactMultiplicity.lean` and `Sheshadri/ConormalBorderMultiplicity.lean`.

What "end to end" means here

For every pair of natural numbers n, m with $n \ge 3$, the development proves the ordinary and symmetric lower bounds from a literal determinant representation, or from literal border membership, respectively. The final theorems do not ask the user to provide a Bézout theorem, a generic flag, a degree formula, a finite section, a specialization principle, or a geometric certificate. These inputs are proved or constructed in the dependency chain.

The formal implementation uses *local Gauss sections and their full intersection lengths* to obtain the numerical consequences of conormal/Gauss specialization. It is not a claim that every

intermediate global-cycle assertion in the main text has been formalized verbatim. Section D.11 specifies this distinction.

D.1 The statements checked by Lean

Throughout the development, the target is the actual multivariate polynomial

$$f_n = \sum_{i\in\{0,\dots,n-1\}} X_i^n \quad \text{in} \quad \mathbb{C}[X_0,\dots,X_{n-1}].$$

This indexing differs from $F_n = x_1^n + \cdots + x_n^n$ only by renaming variables. An affine matrix pencil is an $(n+1)$-tuple of constant $m \times m$ matrices, and its polynomial is literally

$$\det\left(A_0 + \sum_{i=0}^{n-1} X_i A_{i+1}\right).$$

In the symmetric model every coefficient matrix is required to equal its transpose. The definitions occur in `PolynomialModel.lean`, in the namespace `Sheshadri.PolynomialModel`.

Put

$$C(n) = n(n-1)^{n-2},$$

$$B(m,n) = \sum_{i=1}^{n-1} \binom{m}{i}\binom{m-1}{n-1-i}\binom{n-2}{i-1},$$

$$B_{\mathrm{sym}}(m,n) = 2^{n-2}\binom{m}{n-1}.$$

Their Lean names are, respectively, `fermatCount`, `ordinaryBezout`, and `symmetricBezout`. These are natural-number definitions, including the usual zero values of binomial coefficients outside their range. The final real-number inequalities use `Real.exp 1` for e.

The following is a complete account of the inputs of the final endpoints. All entries quantify only $n, m \in \mathbb{N}$, have the additional hypothesis $3 \le n$, and have no further implicit geometric or typeclass parameter. The short names in the first table are in the namespace `Sheshadri.ConormalExactMultiplicity`.

Lean theorem	Remaining representation hypothesis	Conclusion
`ordinary_exact_count`	`ExactRepresentation n m Complex`	$C(n) \le B(m,n)$
`symmetric_exact_count`	`SymmetricExactRepresentation n m Complex`	$C(n) \le B_{\mathrm{sym}}(m,n)$
`ordinary_exact_quadratic`	`ExactRepresentation n m Complex`	$\frac{(n-1)^2}{4e} \le m$
`symmetric_exact_quadratic`	`SymmetricExactRepresentation n m Complex`	$\frac{(n-1)^2}{2e} \le m$

The border names below are in `Sheshadri.ConormalBorderMultiplicity`. In each paired row, "ordinary" and "symmetric" select the corresponding hypothesis and conclusion.

Lean theorem	Remaining border hypothesis	Conclusion
`ordinary_border_count`, `symmetric_border_count`	`OrdinaryFiniteSupportBorder n m`, `SymmetricFiniteSupportBorder n m`	$C(n) \le B(m,n)$ or $B_{\mathrm{sym}}(m,n)$

Lean theorem	Remaining border hypothesis	Conclusion
`ordinary_border_quadratic`, `symmetric_border_quadratic`	The same finite-support predicates	$\frac{(n-1)^2}{4e} \le m$ or $\frac{(n-1)^2}{2e} \le m$
`ordinary_borderAtSize_count`, `symmetric_borderAtSize_count`	`OrdinaryBorderAtSize n m`, `SymmetricBorderAtSize n m`	The corresponding count bound
`ordinary_borderAtSize_quadratic`, `symmetric_borderAtSize_quadratic`	The same canonical-support predicates	The corresponding quadratic bound
`ordinary_zariski_count`, `symmetric_zariski_count`	`OrdinaryZariskiBorder n m`, `SymmetricZariskiBorder n m`	The corresponding count bound
`ordinary_zariski_quadratic`, `symmetric_zariski_quadratic`	The same Zariski-border predicates	The corresponding quadratic bound

For example, the full mathematical content of the ordinary finite-support border endpoint is exactly

$$\forall n, m \in \mathbb{N}, \quad 3 \le n \implies \mathrm{OrdinaryFiniteSupportBorder}(n, m) \implies \frac{((n : \mathbb{R}) - 1)^2}{4 \exp(1)} \le (m : \mathbb{R}).$$

Positive matrix size is derived inside the proof. The stronger count theorems are retained rather than only their asymptotic consequences. The direct Zariski quadratic endpoints compose the Zariski count theorems with `Sheshadri.ordinary_quadratic_bound` and `Sheshadri.symmetric_quadratic_bound`. They do not need a Zariski-to-Euclidean closure theorem.

There is also an exact, non-Fermat version. The theorems `ordinary_smooth_degree_bound` and `symmetric_smooth_degree_bound` assume $n \ge 2$, a nonzero homogeneous f of positive degree d, a literal determinant representation, and

$$x \ne 0, \quad f(x) = 0 \quad \implies \quad \text{some } \partial_i f(x) \ne 0.$$

They conclude $d(d-1)^{n-2} \le B(m, n)$ or $d(d-1)^{n-2} \le B_{\mathrm{sym}}(m, n)$. The smoothness and degree hypotheses are proved for f_n before applying these results.

D.2 The closure definitions are literal

Let S be a finite set of exponent vectors. The coefficient space used in `BorderRepresentation.lean` is the finite-dimensional complex space $\mathbb{C}^S$, equipped with its usual topology. A coefficient vector c denotes the polynomial $\sum_{\alpha \in S} c_\alpha X^\alpha$. The ordinary locus consists of exactly those vectors whose polynomial is a size-m affine determinant; the symmetric locus additionally requires symmetric coefficient matrices. Thus `OrdinaryFiniteSupportBorder n m` means that there exists a finite S and a vector representing f_n that belongs to the Euclidean closure of this actual locus. No family or specialization assertion is part of this definition.

The canonical `AtSize` predicates use `coefficientSupport n m`. The development proves that this finite set contains the support of f_n and of every size-m affine determinant. The theorems `ordinaryLocus_eq_range` and `symmetricLocus_eq_range` identify the loci with the coefficient images of all such pencils. Consequently, the canonical support introduces no unproved restriction on the determinant locus.

The Zariski predicates, defined in `ZariskiBridge.lean`, use the common zeros of all polynomials vanishing on the actual coefficient locus. The proof needed for the Euclidean endpoints is only

$$\overline{T}^{\mathrm{Euclidean}} \subseteq \overline{T}^{\mathrm{Zariski}}.$$

It is proved using continuity of polynomial evaluation. The finite-support coefficient reindexing required to apply this inclusion is also proved. The final argument then constructs its algebraic family directly from Zariski membership. It neither assumes that these two closures agree for arbitrary sets nor uses their reverse inclusion. An older conditional interface named `UniversalPolynomialImageClosure` is retained in the source, but it is not an input to any final endpoint listed above.

D.3 A definition-level dictionary

The formal objects below are concrete algebraic objects, rather than names for desired numerical properties. Local module names in this appendix refer to files in the `Sheshadri/` directory.

Mathematical object	Formal realization
Affine determinant and target	`PolynomialModel.affinePencil`, `determinant`, and `fermat`; actual matrices and multivariate polynomials.
Polar section	`SmoothGaussSectionLength.polarChartIdeal`; the ideal generated by f, the selected derivative combinations, and $a \cdot x - 1$.
Local intersection algebra	`IsolatedLocalMultiplicity.LocalIntersectionRing`; the actual polynomial quotient localized at the evaluation point.
Multiplicity	`IsolatedLocalMultiplicity.multiplicity`; module length of that local ring over itself, with proved equality to field dimension in the rational Artinian setting.
Special Gauss section	`ConormalGaussSection.Section`; actual flags and distinct points, vanishing, finite quotient, Artinian local rings, smoothness, and a proved length sum. `exists_section` constructs the record.
Border family	`DeterminantDvrFamily.Family`; a DVR, residue and generic maps, a polynomial over the DVR, one generic pencil, a uniformizer, and literal specialization and determinant identities.
Open-locus saturation	`HomogeneousSystemSaturation.saturation`; the ideal of q such that, for every displayed g_i, some power $g_i^{N_i} q$ belongs to the original ideal.
Finite generic section	`ActualGenericGaussSection.exists_actual_section`; constructs flags with prescribed residues and a finite quotient by the derivative saturation of the actual generic polar equations.
Incidence multiplicity	The actual normalized kernel-incidence equation quotient in a valid projective chart, localized at its evaluation point; identified with the polar local ring by proved ring equivalences.

For clarity, write the special affine polar ideal as

$$I_0 = \left(f_n,\ \sum_i \ell_{1i}\partial_i f_n, \ldots, \sum_i \ell_{n-2,i}\partial_i f_n,\ a \cdot x - 1 \right).$$

It has n displayed equations in n affine variables. If p is one of its selected points, its contribution is

$$\mu_p = \operatorname{length}_{(\mathbb{C}[x]/I_0)_p}(\mathbb{C}[x]/I_0)_p,$$

not an assigned weight and not merely the number one. This distinction is essential when the polar section is nonreduced.

D.4 Step 1: construct the special Gauss section and its degree

The construction begins with a homogeneous polynomial that is smooth away from the cone vertex. `SmoothGaussPolars` proves the relevant Euler identity and gradient basepoint-freeness. The modules `GenericGaussPolars`, `IrrelevantSaturation`, and `GenericGaussPolarSequence` choose successive polar combinations by avoiding finitely many associated primes. Irrelevant saturation preserves sufficiently high graded pieces, while the chosen equations act regularly on the appropriate saturated quotients. Generic regularity is thus the output of an existence proof, not a premise supplied by the caller.

`SmoothGaussDegree` uses exact sequences of actual graded quotients to calculate the eventual Hilbert value after the $n-2$ polar cuts. It obtains $d(d-1)^{n-2}$. The construction then chooses a linear form avoiding the resulting projective support. Setting this form to one gives an actual finite-dimensional affine algebra, as proved in `SmoothGaussSection` and `SmoothGaussSectionLength`.

The modules `ArtinianLocalDecomposition`, `FiniteQuotientPointFamily`, and `SmoothGaussPointFamily` identify the dimension of this finite algebra with the sum of the full local lengths at its distinct support points. The resulting theorem `ConormalGaussSection.exists_section` supplies all the data in the `Section` record and the identity

$$\sum_p \mu_p = d(d-1)^{n-2}.$$

Specializing to the actual Fermat polynomial gives $\sum_p \mu_p = C(n)$. The endpoint proof invokes this constructor and discharges its hypotheses; it does not assume the existence or the numerical degree of such a section.

The chart condition $a \cdot x = 1$ is chosen after the homogeneous polar section. The proof does not fix an original Fermat coordinate to one and then explicitly solve the resulting polar equations to obtain the lower count. The degree comes from the regular-cut Hilbert calculation.

D.5 Step 2: construct a genuine DVR family from the border

The border argument starts with the coefficient-space membership of Section D.2. `AlgebraicCurveSelection` and `CurveThroughBoundary` construct a curve through the boundary point of the polynomial-image closure. `NormalGoingDown`, `CurveNormalization`, and `FiniteCurveExtension` construct the normalized branch and the finite extension needed to carry a coherent generic preimage. Localizing at the selected point supplies a discrete valuation ring D, a residue map $D \to \mathbb{C}$, and its actual fraction field L.

`GenericPolynomialLift` and `AlgebraicBorderFamily` produce one tuple of generic matrix parameters. They do not choose unrelated representations at separate fibres. Pole clearing in `DvrRationalEntries` uses one common power of a uniformizer. `DeterminantDvrFamily` then reconstructs a polynomial $F \in D[x_1, \dots, x_n]$ and a pencil A over L satisfying the actual polynomial identities

$$F \bmod \pi = f_n, \qquad F \otimes_D L = \det A(x).$$

The theorem `Family.isFractionRing` verifies that the field used by the pencil is the fraction field of the constructed DVR. Symmetry is retained in the symmetric construction.

The final family-selection results are `ordinary_border_exact_or_family` and `symmetric_border_exact_or_family`. They return either an exact representation, to which the exact theorem applies, or the family just described. An arbitrary algebraic curve is never replaced by an assumed parametrization over $\mathbb{C}[t]$, and the generic field L is never identified with $\mathbb{C}$.

D.6 Step 3: prove that a compatible generic section is finite

It would be incorrect to infer finite generic fibre merely from finite special fibre: components can escape the special fibre. The development therefore proves generic finiteness separately, using the actual family.

`DeterminantDvrHomogeneous` homogenizes F in an added variable and proves the dehomogenization and derivative identities. Its theorem `generic_hypersurface_hilbert` bounds the degree of the initial Hilbert polynomial by $n-1$. The saturated generic construction works on the open locus where the added coordinate and at least one spatial derivative are nonzero.

The prescribed special flags cannot simply be replaced by unrelated generic flags. `RegularFlagsWithResidue.exists_regular_coefficients` chooses coefficients in D with exactly the required residues, while avoiding the finitely many associated primes of the generic quotients. The choices are perturbations by uniformizer multiples. The modules `PrescribedResidueGaussSequence` and `FinitePrescribedGaussSection` apply this construction to all $n-2$ polar cuts and the final primal cut. Normalization of the primal equation divides only by a proved unit of D and preserves its prescribed residue.

`HomogeneousHilbertDegree` proves the degree decrease under a regular homogeneous cut and the required monotonicity under enlargement of the ideal. `SaturatedSectionHilbert` applies these results through the actual cut sequence. After $n-1$ cuts the Hilbert function is eventually constant. Regularity of the added coordinate then proves that the corresponding affine chart algebra is finite-dimensional.

Dehomogenization identifies an actual ideal $J \subseteq L[x]$. If

$$I_\eta = \left(\det A,\ \sum_i \widetilde{\ell}_{1i}\partial_i \det A, \ldots, \sum_i \widetilde{\ell}_{n-2,i}\partial_i \det A,\ \widetilde{a}\cdot x - 1\right),$$

then `ActualGenericGaussSection.exists_actual_section` proves

$$\dim_L L[x]/J < \infty,$$
$$I_\eta \subseteq J \subseteq \mathrm{Sat}_{\partial \det A}(I_\eta), \qquad J = \mathrm{Sat}_{\partial \det A}(I_\eta).$$

Here $q \in \mathrm{Sat}_g(I)$ means $\forall i\, \exists N_i \geq 0 :\ g_i^{N_i} q \in I$. This operation removes components supported on the excluded closed locus; it does not take a radical. Where one g_i is a unit, the localized saturated and original ideals are proved equal.

Finally, `SaturatedChartSmoothness` constructs a derivative combination whose class in the full finite quotient $L[x]/J$ is a unit. Regularity gives injectivity of multiplication; finite dimension gives surjectivity and hence the inverse. Therefore all geometric support points of J are smooth points of the determinant hypersurface, including after passage to an algebraic closure. The full quotient by I_η need not be finite: excess components outside this smooth section are allowed throughout.

D.7 Step 4: preserve singular local multiplicities

The lower comparison must preserve nilpotents, not merely show that the selected special points are in a limiting support. This is the role of `DVRRootPersistence`.

At a selected special point, the actual local polar algebra is Artinian and is given by a square equation system. The regular-sequence machinery in `CoordinatePowerRegular`, `LocalCompleteIntersection`, and `IsolatedCompleteIntersection` derives regularity of these equations in the local polynomial ring. It does not require a nonzero Jacobian determinant. The total local equation quotient then has regular uniformizer action. Cancelling generic denominators shows that contraction from the generic fibre annihilates the entire original special local algebra.

`DVRSaturatedPersistence` handles the additional saturation in J. At a selected special point, one spatial derivative has nonzero value, and consequently is a unit in its full Artinian local ring. Power membership in the saturation, followed by cancellation of this unit, shows that every contracted element of J still annihilates that ring. `DVRSaturatedGaussPersistence` applies this result to the actual family and the residue-preserving flags. Its theorem `section_saturated_contraction_local_eq` proves equality of the corresponding localized ideals. Thus this step retains local scheme structure and local length.

Let $T = J \cap D[x]$ and let T_0 be its residue ideal. Simultaneous interpolation onto the full special local algebras gives

$$\sum_p \mu_p \;\le\; \dim_{\mathbb{C}} \mathbb{C}[x]/T_0.$$

The quotient $D[x]/T$ embeds in its generic fibre and is torsion-free over D. `DVRFibreDimension` proves the residue-dimension inequality

$$\dim_{\mathbb{C}} \mathbb{C}[x]/T_0 \;\le\; \dim_L L[x]/J.$$

It does not assume finite generation of the whole horizontal algebra over D; the already proved finite generic dimension is used where needed. `DVRSectionLength.selected_lengths_le_generic` combines these facts, and `DVRGaussLengthBound.family_section_degree_le` instantiates them to obtain

$$n(n-1)^{n-2} \;\le\; \dim_L L[x]/J. \tag{9}$$

D.8 Step 5: Bézout with isolated singular zeros and excess

The precise Bézout statement matters. The development proves an isolated-intersection inequality; it does not assert that an arbitrary nonproper intersection has the expected total length. The theorem `FieldSingularBezoutWithExcess.isolated_multiplicity_bound` is over a characteristic-zero field and has the following inputs:

(1) Three finite nonempty homogeneous coordinate blocks of projective dimensions a, b, c, and exactly $a + b + c$ actual equations.
(2) The displayed multihomogeneity of each equation, with multidegree $(d_{e,1}, d_{e,2}, d_{e,3})$.
(3) A finite family of common zeros, each with a specified nonzero chart coordinate in every block, distinct modulo the independent projective block scalings.
(4) Artinianness of each selected point's actual localized chart equation ring.

Its conclusion is

$$\sum_p \operatorname{length}(\mathcal{A}_p) \;\le\; [H_1^a H_2^b H_3^c] \prod_e (d_{e,1}H_1 + d_{e,2}H_2 + d_{e,3}H_3). \tag{10}$$

The family of selected zeros need not exhaust the zero set. Other components may have positive dimension. The selected roots need not be simple or reduced. A companion theorem, `isolated_multiplicity_bound_of_isOpen`, derives Artinianness from literal isolatedness expressed as an open singleton in the actual chart spectrum.

The proof of (10) is itself part of the formalization. It interpolates the given equations with a split comparison system of the same multidegrees, contracts the generic ideal without radicalizing it, and proves flatness and conservation of actual finite multigraded pieces. Local complete-intersection persistence retains all selected local lengths at the original endpoint. At the comparison endpoint, the split equation quotient surjects onto the contracted-fibre quotient, giving the upper-bound inequality in the correct direction.

The split-system estimate uses a Hilbert-function inequality for products, not an unproved regular-sequence equality in the entire homogeneous coordinate ring. Expanding products gives finitely

many branches. Branches selecting too many linear equations in a block vanish in sufficiently positive degree; branches meeting exactly the dimension quotas contribute at most one. The number of quota choices is the coefficient on the right of (10). All required degree thresholds and componentwise shifts are proved. This approach allows irrelevant and excess components without identifying point counts with the degree of an arbitrary homogeneous quotient.

The main dependency modules for this part are `FieldSingularBezoutDeformationBound`, `FieldBezoutSplitHilbert`, `FieldBezoutVariableTransport`, and `FieldSingularBezoutWithExcess`, together with their local persistence and multigraded algebra dependencies. The comparison split system is a general Bézout upper-bound construction; it is unrelated to a fixed-coordinate explicit count for the Fermat polynomial.

D.9 Step 6: apply Bézout to the actual determinant incidence

At a smooth determinant zero, the proof constructs normalized kernel vectors and valid cofactor charts. In the ordinary model it also constructs one reduction matrix valid simultaneously for the finite selected family. The Schur-complement calculations are equalities of ideals and actual quotient/local-ring equivalences, not only equalities of their zero sets. The symmetric model uses its native one-kernel-vector incidence rather than discarding the symmetry condition.

`FieldPolarMultiplicityBound` applies the singular isolated-zero Bézout theorem to these literal incidence equations. It proves the equation count, multidegrees, projective distinctness and Artinian local conditions from the constructed polar data. The local equivalences transfer the full intersection lengths. The resulting incidence coefficients are precisely $B(m,n)$ and $B_{\mathrm{sym}}(m,n)$.

The generic field of the border family need not be algebraically closed. `ArbitraryFieldFinitePolarBound` therefore extends the actual finite algebra $L[x]/J$ to an algebraic closure $\overline{L}$. Explicit tensor/quotient equivalences prove

$$\dim_{\overline{L}}\left(\overline{L}\otimes_L L[x]/J\right) = \dim_L L[x]/J.$$

The unit derivative combination remains a unit after extension, so every support point over $\overline{L}$ is smooth. The proved saturation inclusions suffice to identify each localized J-algebra with the original polar algebra. The argument uses the required inclusion under mapping an ideal; it does not assume that saturation commutes with every base change.

`FieldFinitePolarBound` constructs the complete point family of the finite quotient and sums its local lengths. Consequently `ArbitraryFieldFinitePolarBound.ordinary_finite_saturated_polar_bound` and its symmetric counterpart give

$$\dim_L L[x]/J \le B(m,n), \qquad \text{or} \qquad \dim_L L[x]/J \le B_{\mathrm{sym}}(m,n). \tag{11}$$

No generic homogeneity of the original affine determinant polynomial and no finite-dimensionality of its entire unsaturated polar quotient is assumed here.

D.10 Step 7: assemble the endpoints and the constants

The family-level theorem `ConormalBorderMultiplicity.ordinary_family_count` constructs the special section and the compatible finite generic section, applies (9) and (11), and concludes $C(n) \le B(m,n)$. The symmetric theorem performs the same assembly with the symmetric incidence. The Zariski endpoints invoke the exact-or-family constructor, and the Euclidean endpoints invoke the proved one-way closure transfer. Thus the intermediate family and section records are outputs of proved constructors, not undeclared obligations of the endpoint user.

`Bezout.lean` proves the coefficient identities and the Vandermonde estimate. `Numerical.lean` proves

$$B(m,n) \le \left(\frac{4em}{n-1}\right)^{n-1}, \qquad B_{\mathrm{sym}}(m,n) \le \left(\frac{2em}{n-1}\right)^{n-1}.$$

Combining these with $(n-1)^{n-1} \le C(n)$ and the positivity following from $n \ge 3$ yields the two quadratic bounds. All casts between natural numbers and real numbers, the exponential constants, and the power comparison are checked by Lean.

The exact proof is shorter: it constructs the special section, identifies its actual local rings with the kernel incidence, and applies Bézout directly. It does not need the border family or specialization comparison.

D.11 What the formalization establishes, and what it does not

The verified numerical chain for the border theorem is

$$\underbrace{n(n-1)^{n-2}}_{\text{constructed special local lengths}} \le \underbrace{\dim_L L[x]/J}_{\text{constructed generic section}} \le \underbrace{B(m,n)}_{\text{proved incidence Bézout coefficient}},$$

with the symmetric replacement in the last term. This is a local Gauss-section implementation of the conormal/Gauss specialization argument. It establishes the stated exact and border lower-bound endpoints without assuming a global projective cycle/intersection certificate.

The written proof also distinguishes numerical constancy of fibre multidegrees from rational equivalence of fibres over an arbitrary base curve: the relevant numerical constancy is obtained from the bigraded Euler–Hilbert calculation. Likewise, the cone comparison is made away from the vertex, and the primal hyperplane is chosen to avoid the vertex. No global projective-line-bundle description through the vertex is needed. These distinctions are consistent with the local-section implementation; the end-to-end Lean guarantee remains the precise endpoint statements in Section D.1.

The development also contains separately proved results on actual Gauss families, flatness, conormal containment, grading, Hilbert functions, and projective charts. It does not claim a separately complete global Chow ring library, the full global projective fibre-cycle assembly of the main text, or an identification of every concrete local chart ring with a stalk on a product of Mathlib `Proj` schemes. Those broader claims are not needed as hypotheses of the assembled endpoints.

Older modules retain conditional interfaces called `GaussDegreeTheory` and `FamilyCertificate`. A theorem accepting one of these records remains conditional. Importing its module does not make the record an axiom, but a clean axiom list also does not prove the record exists. The final `Multiplicity` endpoint modules use the constructed local-section route described above and do not require those interfaces.

The only affine chart used in the generic construction sets the *added homogenizing coordinate* x_0 to one. No original coordinate of F_n is set to one, and no explicit count of polar points is used: the lower count comes from the Hilbert-function calculation of Section D.4.

D.12 What the audits check

The release's entry-point dependency graph contains 225 local modules. The principal axiom audit requests 349 reports. The accompanying `BUILD_LOG.txt`, `AXIOM_AUDIT.txt`, and `IMPORT_AUDIT.txt` record successful build and audit results. The scripts recompute the checks from the delivered source; these counts are descriptive release metadata rather than mathematical assumptions. The implementation and source-level review were AI-assisted. The semantic audit documents record

separate automated source reviews, not independent human peer review. The compiler and axiom transcripts are the reproducible machine-checking evidence.

There are three distinct checks.

(1) **Compilation.** Lean elaborates the local development against the pinned dependencies and checks the resulting proof terms. A build failure is not converted into a successful verification claim.
(2) **Foundational dependencies.** `AxiomAudit.lean` runs `##print axioms` on the principal declarations, including the final endpoints. The only reported axioms are Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. `audit_axioms.py` checks that every requested report is present and rejects unexpected dependencies. `audit_imports.py` checks the local entry-point import graph for missing files, custom axioms, unfinished proofs, unsafe shortcuts and the excluded alternative explicit-slice modules.
(3) **Mathematical statement and applicability.** `END_TO_END_HYPOTHESES_AUDIT.txt` reviews the literal definitions, the final theorem types, and how the proof supplies every constructor's applicability conditions. This check is necessary because a theorem of the form "if the desired count bound holds, then the desired conclusion holds" can compile with only standard foundational axioms while proving little about the intended geometric problem.

These checks address different questions. A source scan is not a substitute for compilation, an axiom report is not a semantic audit, and the existence of a successful transcript is not a substitute for rebuilding the sources when independent reproduction is desired. The trusted base includes the Lean kernel and the pinned mathematical definitions and dependencies. The verification is a proof for all admitted n, m, not numerical testing of finitely many dimensions.

Construction-specific reading guides are `GAUSS_DEGREE_CONSTRUCTION_AUDIT.txt`, `BORDER_DVR_CONSTRUCTION_AUDIT.txt`, `GENERIC_GAUSS_SECTION_AUDIT.txt`, `DVR_LOCAL_PERSISTENCE_AUDIT.txt`, and `EXACT_MULTIPLICITY_AUDIT.txt`. Historical audit files are retained to document earlier stages; their earlier lists of remaining obligations are superseded by `ORIGINAL_CONORMAL_SCOPE.txt` and `END_TO_END_HYPOTHESES_AUDIT.txt`.

D.13 Reproducing the build from the release sources

The release is source-complete for its local entry-point graph. It does not vendor the Lean compiler, Mathlib, or precompiled Mathlib caches. A fresh machine therefore needs an Internet connection for the initial toolchain and dependency download, Git, Python 3, a shell capable of running the supplied Bash script, and Lean's toolchain manager Elan. The mathematical toolchain is pinned to

- Lean `leanprover/lean4:v4.19.0`;
- Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` (Mathlib v4.19.0);
- the remaining dependency revisions in the delivered `lake-manifest.json`.

Download and extract the archived release, or clone the repository (both linked at the start of this appendix), and keep its manifest. From the directory containing `lean-toolchain`, `lakefile.toml`, and `verify.sh`, run:

```
elan toolchain install leanprover/lean4:v4.19.0
lake env lean --version
lake exe cache get
./verify.sh
```

If the extraction did not preserve executable permissions, use `bash verify.sh`. The script executes, in order:

```
lake build
lake env lean AxiomAudit.lean | tee AXIOM_AUDIT.txt
```

```
python3 audit_axioms.py
python3 audit_imports.py
```

It stops on failure. A successful return code is zero. The expected audit summaries report all requested declarations, only the standard foundational axioms, and a complete entry-point source graph with no prohibited shortcuts. The Mathlib cache is a build accelerator; the release supplies source proofs for its own modules. A reader wishing to rebuild dependencies from source may omit the cache download and allow the dependency build to complete instead.

The manifest-preserving build above is the reproduction procedure for the archived release. `lake update` is useful when deliberately regenerating dependency metadata, but is not necessary for this procedure and may change transitive dependency resolutions. If it is used, retain a copy of the archived manifest, compare every resolved revision with that copy, and restore the archived manifest before claiming an exact release reproduction. The Mathlib checkout can be checked directly:

```
git -C .lake/packages/mathlib rev-parse HEAD
```

It must print the Mathlib commit above. The release's `SOURCE_SHA256SUMS.txt` records the delivered file contents except the checksum manifest itself. Verify these checksums on the untouched extraction before rerunning scripts that regenerate logs. On a system with the GNU checksum utility, run `sha256sum -c SOURCE_SHA256SUMS.txt` from the directory to which the manifest paths are relative.

`BUILD_ENVIRONMENT.txt` describes the original build environment and the small executable-location compatibility wrapper needed by that container. The wrapper changes only the process executable-location lookup and is not part of the mathematical development. A normal Lean installation does not need it. The compiler and shared-library hashes are recorded there for inspection.

D.14 A short independent endpoint inspection

A reader need not begin by reading every implementation file. After building the release, create a new file named `EndpointInspection.lean` in the project root with the following contents. This file only inspects the already defined statements; it does not introduce assumptions or replace a proof.

```
import OriginalConormal

set_option pp.explicit true
set_option pp.universes true

open Sheshadri.ConormalExactMultiplicity in
#check @ordinary_exact_quadratic
open Sheshadri.ConormalExactMultiplicity in
#check @symmetric_exact_quadratic

open Sheshadri.ConormalBorderMultiplicity in
#check @ordinary_border_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#check @symmetric_border_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#check @ordinary_borderAtSize_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#check @symmetric_borderAtSize_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#check @ordinary_zariski_count
open Sheshadri.ConormalBorderMultiplicity in
```

```
#check @symmetric_zariski_count
open Sheshadri.ConormalBorderMultiplicity in
#check @ordinary_zariski_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#check @symmetric_zariski_quadratic

open Sheshadri.ConormalBorderMultiplicity in
#print axioms ordinary_border_quadratic
open Sheshadri.ConormalBorderMultiplicity in
#print axioms symmetric_border_quadratic

#print Sheshadri.PolynomialModel.ExactRepresentation
#print Sheshadri.PolynomialModel.SymmetricExactRepresentation
open Sheshadri.BorderRepresentation in
#print OrdinaryFiniteSupportBorder
open Sheshadri.BorderRepresentation in
#print OrdinaryBorder
open Sheshadri.BorderRepresentation in
#print ordinaryLocus
open Sheshadri.ZariskiBridge in
#print OrdinaryZariskiBorder
open Sheshadri.ZariskiBridge in
#print zariskiClosure
```

Run `lake env lean EndpointInspection.lean`. The output makes all implicit parameters visible. In particular, the final exact and border statements must have only the natural numbers n, m, the condition $3 \leq n$, and the literal representation or border predicate. The ordinary and symmetric endpoint axiom lists must contain no custom axiom or `sorryAx`. Repeat the definition inspection with the symmetric border names if desired; their extra condition is literal matrix symmetry.

For a mathematical review, a useful order is: inspect the definitions and final types; read the two endpoint modules; follow `exists_section` and `exists_actual_section`; inspect `family_section_degree_le`; then inspect the arbitrary-field finite polar bound and the singular Bézout statement. At each step, check both the intermediate theorem's assumptions and the constructor that supplies them. This follows the actual proof dependency chain and separates a reusable conditional lemma from an assembled unconditional endpoint.

References

[1] J. Alper, T. Bogart, and M. Velasco. A lower bound for the determinantal complexity of a hypersurface. *Found. Comput. Math.* 17(3):829–836, 2017.

[2] P. Bürgisser. *Completeness and Reduction in Algebraic Complexity Theory.* Algorithms and Computation in Mathematics 7, Springer, 2000.

[3] W. Fulton. *Intersection Theory*, 2nd ed. Springer, 1998.

[4] I. M. Gelfand, M. M. Kapranov, and A. V. Zelevinsky. *Discriminants, Resultants, and Multidimensional Determinants.* Birkhäuser, 1994.

[5] R. Hartshorne. *Algebraic Geometry.* Graduate Texts in Mathematics 52, Springer, 1977.

[6] S. L. Kleiman. The transversality of a general translate. *Compositio Math.* 28:287–297, 1974.

[7] M. Kumar and B. L. Volk. A lower bound on determinantal complexity. In *36th Computational Complexity Conference (CCC 2021)*, LIPIcs 200, art. 4. Journal version: *Comput. Complexity* 31 (2022), art. 8. Full version: *arXiv:2009.02452.*

[8] M. Kumar and B. L. Volk. A quadratic lower bound on determinantal complexity. *arXiv:2609.34462*, 2026.

[9] J. M. Landsberg, L. Manivel, and N. Ressayre. Hypersurfaces with degenerate duals and the geometric complexity theory program. *Comment. Math. Helv.* 88(2):469–484, 2013. See also *arXiv:1004.4802.*

[10] T. Mignon and N. Ressayre. A quadratic bound for the determinant and permanent problem. *Int. Math. Res. Not.* 2004(79):4241–4253, 2004.

[11] D. Mumford. *The Red Book of Varieties and Schemes*, 2nd expanded ed. Lecture Notes in Mathematics 1358, Springer, 1999.
[12] R. Piene. Polar classes of singular varieties. *Ann. Sci. Éc. Norm. Supér.* 11(2):247–276, 1978.
[13] K. Sheshadri. A near-quadratic lower bound on the determinantal complexity of $\sum_i x_i^n$ via the polar degree of the tangent cone. Unpublished manuscript, 2026.
[14] K. Sheshadri. A symmetric determinantal lower bound for diagonal power sums via polar degree. *arXiv:2606.11090*, 2026.
[15] A. J. Sommese and C. W. Wampler. *The Numerical Solution of Systems of Polynomials Arising in Engineering and Science*. World Scientific, 2005.
[16] E. A. Tevelev. *Projective Duality and Homogeneous Spaces*. Encyclopaedia of Mathematical Sciences 133, Springer, 2005.
[17] The Stacks Project Authors. *The Stacks Project*. Tags 0B9T (Euler characteristic in proper flat families), 0BEM–0BES (numerical intersections and generic-length multiplicities). https://stacks.math.columbia.edu.